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Roll No. 22UMEC1134

BE II (ME)
3

Engg. Thermo.

SECOND B.E. EXAMINATION, 2023

(MECHANICAL ENGINEERING)

(III SEMESTER CBCS)

ME – 201 A: ENGINEERING THERMODYNAMICS (M)

Time – Three Hours

Maximum Marks – 80

Note: (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) In a reciprocating engine, the mass of gas occupying the clearance volume is m_c kg at state p_1, u_1, v_1, and h_1. By opening the inlet valve, m_f kg of gas is taken into the cylinder, and at the conclusion of the intake process the state of the gas is given by p_2, u_2, v_2, and h_2. The state of the gas in the supply pipe is constant and is given by p_p, u_p, v_p, h_p and V_p. How much heat is transferred between the gas and the cylinder walls during the intake process? 8

   (b) State the Kelvin- Planck statement of second law of thermodynamics. Write down the corollaries of second law of thermodynamics. 8

   PYQ 2019

2. (a) Derive the expression of exergy for a steady flow system (open system). 8

   (b) 4 kg of water at 50°C are mixed with 5 kg of water at 100°C in a steady flow process. Calculate:
   (i) the temperature of resulting mixture
   (ii) the change in entropy
   (iii) the unavailable energy with respect to the energy receiving water at 50°C. 8

cont.

3. (a) Derive the vander Waals constants in terms of critical properties.

PYQ 2017

$$a = 3p_c.v_c^2,\ b = \frac{1}{3}v_c\ \&\ R = \frac{8a}{9T_c v_c}$$ 8

(b) State Dalton's law of partial pressure, Amagat's law of additive volumes and gibbs theorem. 8

PYQ 2019

4 a) Write down the first and second Tds equations. Derive the expression for the difference in heat capacities, C_p & C_v. What does the expression signify? 8

PYQ 2019

(b) Explain joule kelvin effect using T-p diagram. Mention inversion curve, cooling and heating region on the T-p diagram. Derive the value of joule kelvin coefficient for ideal gas. 8

PYQ 2019

5. (a) Derive the expression of work done per cycle for single stage reciprocating air compressor. 8

PYQ

(b) A multi stage air compressor takes in air at 1 bar, 298 K and delivers at 36 bar. The maximum temperature in any stage is not to exceed 390 K. If the law of compression and expansion is $pv^{1.3}$ = constant, find the number of stages for minimum power input. Estimate the power required. 8

PYQ 2019

6. (a) What is stoichiometric air fuel ratio and excess air? A fuel having formula C_8H_{18} is burnt with chemically correct amount of air at atmospheric pressure. Write the reaction equation and find the stoichiometric air fuel ratio. 8

PYQ 2019

(b) Describe the orsat apparatus with a sketch. Explain how the gases are analysed with the help of this apparatus. 8

PYQ 2019

7. Write short notes on any four- 4x4=16

(i) International practical temperature scale.

(ii) Classification of air compressors → PYQ 2019

(iii) Second law efficiency

(iv) Carnot cycle → PYQ 2018

(v) Effect of pressure ratio and clearance on volumetric efficiency of reciprocating compressor

(vi) Maxwell's equations. PYQ → 2019, 2018

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Roll No. 22UMEC1134

BE II (ME)

3

Mat. Tech.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 202 A: Materials Technology (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Discuss imperfection in crystals with proper diagram. What are the advantages and disadvantage of imperfections. 8

(b) Find out coordination number, atomic radius, no. of atoms in B.C.C., F.C.C. and H.C.P. structures. 6

(c) Construct (101) and (110) plane with in unit cell. 6

2. (a) What are various types of dislocations in two dimension and three dimensions. 8

(b) How dislocations are found out with the help of Berger Vector? 6

(c) What are the cold and hot deformations. What factors responsible to affect them? 6

3. (a) Draw a neat iron carbon equillibirium diagram and label various regions of eutectic and eutecloide phase. 12

(b) In continuation of C.I. equillibirium explain various structure so obtained with their mechanical properties. 8

4. (a) What do you understand by Heat treatment of steel? Why it is performed? What are the advantages and applications in various engineering skills.

(b) Explain Annealing process. Discuss various types of Annealing methods.

5. (a) How hardness is imparted in steel. Explain various Quenching methods and Quenching media. 10

(b) Discuss various Alloying elements in steel in brief. Write uses and applications of various types of carbon steel. 10

6. (a) Discuss various powder manufacturing methods. 10

(b) How Sintering is performed? What are the advantages of sintering? Compare polymers, ceramics & composites. 10

7. Write short notes on (any four): 5X4

(a) Mechanical properties of materials

(b) Fatigue and fracture

(c) Nitriding and cyaniding

(d) Brass, Bronze and Bearing material

(e) Smart materials and its applications

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Roll No. 22.UME.C.1124

B.E. II (ME)

4

Kine. of Mach.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 203 A: Kinematics of Machines - (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Q. No. 1 is compulsory & complete solution (including calculations) for Q. No. 1 is to be given in the drawing sheet.

(iii) All questions carry equal marks.

(iv) Assume suitable data, if not given and mention it clearly.

(v) Draw neat sketches, wherever necessary.

1. (a) Locate all the instantaneous centres of the slider crank mechanism as shown in Fig. 1. The length of crank OB=150 mm and connecting rod AB=450 mm, crank OB makes angle 45° with OA. Point C is located on connecting rod AB at 150 mm from point B. If the crank rotates clockwise with an

angular velocity of 10 rad/second. Calculate the linear velocity of slider A, connecting rod AB and point C located on connecting rod AB. Also calculate the angular velocity and acceleration of the connecting rod AB. 16

2. (a) Write and draw all the possible the inversions of single-slider crank chain mechanism. 8

(b) Distinguish between lower and higher pairs with examples and neat sketches. Explain various lower pairs with sketches. 8

3. (a) In a crank and slotted lever quick return motion mechanism, the distance between the fixed centres is 300 mm and length of driving crank is 150 mm. Find the inclination of the slotted bar with the vertical in the extreme position and time ratio of cutting stroke and to the return stroke. 8

PYQ 2019

(b) Describe Grashof's law and Grubler's law. State how is Grashof's law helpful in classifying the four-bar mechanisms into different types. 8

PYQ 2019 2017,

4. (a) What is uniform pressure and uniform wear theories? Deduce expression for the friction torque considering both the theories for a flat collar. 8

PYQ 2017

(b) A screw jack raises a load of 16 kN through a distance of 150 mm. The mean diameter and the pitch of the screw are 56 mm and 10 mm respectively. Determine the work done and the efficiency of the screw jack when the:

Case 1. Load rotates with the screw.

Case 2. Loose head on which the load rests does not rotate with the screw and the outside and the inside diameters of the bearing surface of the loose head are 50 mm and 10 mm respectively. Take coefficient of friction for the screw and the bearing surface as 0.11. 8

5. (a) Define and elaborate the law of belting. Derive the relation for ratio of belt tensions in a flat-belt drive. 8

PYQ

(b) In a belt drive, the mass of the belt is 1.5 kg/m length and its speed is 6 m/second. The drive transmits 12 kW of power. Determine the initial tension in the belt and the strength of the belt. The coefficient of friction is 0.25 and the angle of lap is 165°. 8

6. (a) A rope drive is required to transmit 250 kW from a sheave of 1 m diameter running at 450 rpm. The safe pull in each rope is 800 N and the mass of the

rope is 0.46 kg/m length. The angle of lap is 160 degree and the groove angle is 45 degree. If the coefficient of friction is between the rope and the sheave is 0.3; calculate the number of rope. 8

(b) Explain the terms slip and creep as referred to belt drive. On what factors does it depend? Explain how creep affects the power transmission capacity of the belt drive. 8

(a) Explain watt's mechanism with neat sketch. Show, how can draw an approximate straight line? 8

(b) A Hook's joint connects two shafts whose axes intersect at 20°. The driving shaft rotates at a uniform speed of 300 rpm. The driven shaft with attached masses has a mass of 60 kg and radius of gyration of 120 mm. Determine the:

(a) Torque required at the driving shafts if a steady torque of 180 Nm resists rotation of the driven shaft and the angle of rotation is 45°.

(b) Angle between the shafts at which the total fluctuation of speed of the driven shaft is limited to 25 rpm. 8

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Roll No. 22UMEC.2134

B.E. II (ME)

3 Mech. of Soli.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 204 A: Mechanics of Solids (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) A load of 2 MN is applied on a short concrete column 500 mm × 500 mm. The column is having 4 steel bars of 10 mm diameter one at each corner. Find the stresses in the concrete and steel bars. Young's modulus for steel is 2×10^5 N/mm^2 and for Concrete is 1.4×10^4 N/mm^2. 8

(b) If the tension test bar is found to taper from (D+a) to (D-a) diameter, prove the error involved in using mean diameter to calculate Young's modulus is (10a/D) percent. 8

2. Draw SFD and BMD for the beam shown in Fig. 1. Also find point of contraflexure.

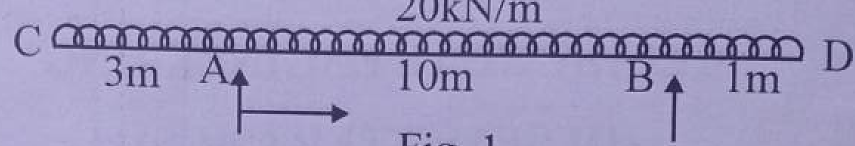

Fig. 1

16

3. (a) Explain theory of simple bending and derive bending equation stating the assumptions. 12

(b) Why the I-section is better than other sections for a beam? 4

4. A simply supported beam is loaded as shown in Fig 2. Determine the deflection at points C and D in the beam. Take $E = 2\times10^5 N/mm^2$ and $I = 2\times10^8 mm^4$.

PYQ-19

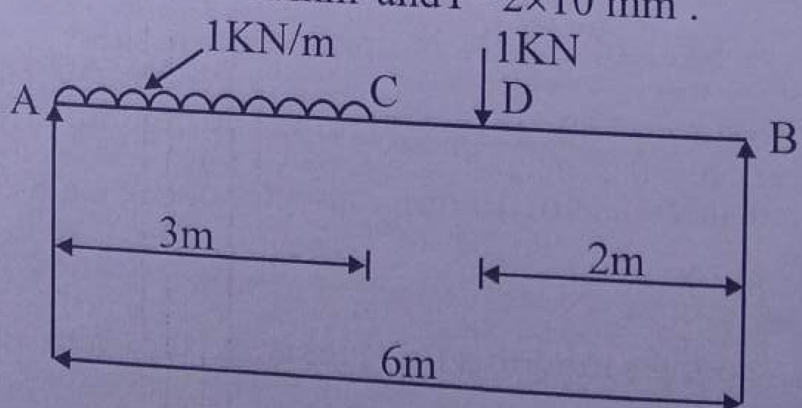

Fig. 2

16

PYQ 5-19 (a) Derive torsion equation stating the assumptions. 8

(b) A bar of steel has rectangular cross section 30×20 mm. Find the percentage decrease of area of cross section, when it is subjected to a tensile force of 120 kN in the direction of its length. Take $E = 2 \times 10^5 N/mm^2$ and $\mu = 0.3$. 8

6. (a) A steel bar of rectangular section 30 mm × 40 mm one end fixed and other is free at each end is subjected to axial compression. The bar is 1.75 m long. Determine the buckling load and the corresponding axial stress using Euler's formula. Determine the minimum length for which Euler's equation may be used to determine the buckling load, if the proportional limit of the material is 200 N/mm^2. Take $E = 2 \times 10^5 N/mm^2$ 16

PYQ 19

PYQ-19

7. (a) At a point in a bracket the stresses on two mutually perpendicular planes are 240 N/mm^2 tensile and 120 N/mm^2 tensile. The shear stress across these planes is 60 N/mm^2. Find the principal stresses and maximum shear stress at the point. 12

(b) Explain limitation of Euler's theory of column. 4

Roll No. 22UMEC1134

B.E. II (ME)

3

Maths.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 205 A: Mathematics (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Solve $\frac{dx}{dt} - 7x + y = 0; \quad \frac{dy}{dt} - 2x - 5y = 0$ 8

-7x, 5x

(b) Solve $(2x^2 + 2xy + 2xz^2 + 1)\,dx + dy + 2z\,dz = 0$ 8

2. (a) Solve the following partial diff. equation by using Charpit's Method

PYQ-2018

(2y-z ; z-t)

$$2xz - px^2 - 2qxy + pq = 0$$ 8

(b) Solve the following partial differential equation by using Monge's method

PYQ 19 approximate

$$x^2 r - 2xs + t + q = 0$$ 8

3. (a) The area A of a circle of diameter d is given for the following values. 8

d :	80	85	90	95	100
A:	5026	5674	6362	7088	7854

Find approximate values for the areas of circle of diameter 82. 5280.017

(b) Use stirling formula to find Y_{28}, given $Y_{20} = 49225$, $Y_{25} = 48316$, $Y_{30} = 47236$, $Y_{35} = 45926$, $Y_{40} = 44306$ 8 47,487.2

4. (a) Use Simpson's 1/3 and 3/8 rule to evaluate the following : 8

PYQ 2018

$$\int_0^1 \frac{dx}{1+x^2}$$

Hence obtain the approximate value of π in each case.

(b) Find the root of the equation $2x - \log_{10} x = 7$ which lies between 3.5 and 4 by Regula falsi method. 8

5. (a) Prove that $L\left\{\frac{\text{Sin}\, t}{t}\right\} = \tan^{-1}\left(\frac{1}{s}\right)$

classwork

and hence find $L\left\{\frac{\text{Sin}\, at}{t}\right\}$ $\frac{a}{s^2+a^2}$ 8

(b) Solve by using Laplace transform 8

PYQ-2017

$$(D^2+9)\, y = \text{Cos}\, 2t$$

given that $y(0) = 1$, $y(\pi/2) = -1$

6. (a) Find the Fourier Sine and Cosine transform of 8

$$f(x) = \begin{cases} 1 & ; \ 0 < x < a \\ 0 & ; \ x > a \end{cases}$$

(b) Find the Fourier Sine transform of e^{-x} ; $x>0$ 8

Hence show that $$\int_0^{\infty} \frac{x \operatorname{Sin} mx}{1+x^2}\, dx = \frac{\pi}{2} e^{-m}$$

7. (a) Find the value of f(5) from the following table by using Lagrange's interpolation formula 8

x:	1	2	3	4	7
f(x):	2	4	8	16	128

(b) Using Runge - Kutta method of fourth order, find an approximate value of y(0.2) and y(0.4) by solving

$$\frac{dy}{dx} = -2xy^2 \text{ with } x = 0,\ y = 1 \text{ ; h being } 0.2$$ 8

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Roll No. ....

BE II (ME)
3

Mech. Meas. & Inst.

SECOND B.E. EXAMINATION, 2023

(MECHANICAL ENGINEERING)

(III SEMESTER CBCS)

ME-206 A: MECHANICAL MEASUREMENTS AND INSTRUMENTATION (M)

Time – Three Hours

Maximum Marks – 80

Note: (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1\. (a) What is a transducer, discuss its various types and desired characteristic. 6

PYQ

(b) Explain Null and deflection methods of measurement. 4

(c) Draw the block diagram of a generalised measurement system. Identify the various elements and point out the function performed by each element. 6

2\. (a) Explain the term impedance loading and matching of a generalised measurement system. Derive the expression for transmitting maximum power from the transducer to the external load. 8

(b) The following readings are taken of a certain are physical length with the help of a micrometer Screw.

PYQ 2019

1.41 1.45 1.53 1.54 1:49 1.51 1.60 1.55 1.47 and 1.66 mm

Assuming that only random errors are present. Calculate the arithmetic mean, the average deviation, standard deviation, variance and the probable error of the reading. 8

Cont.

3. (a) Sketch and explain a thermocouple vaccum gauge with its merits and demerits. 8

2013,19

(b) A Wheatstone bridge circuit comprises four resistance; One resistance is connected in each limb the bridge. Under null mode operation, the solution of the unknown resistance is

$$R_u = \frac{R_2 \, x \, R_3}{R_1}$$

The known resistance and the limiting errors associated with them are:

$R_1 = (100 \pm 0.75\%)$ ohm

$R_2 = (1000 \pm 0.5\%)$ ohm

$R_3 = (850 \pm 0.6\%)$ ohm

Determine the magnitude of unknowns resistance K_u and its limiting errors. Also workout the guaranteed value of the resistance. 8

4. (a) Sketch a typical optical pyrometer. Explain its working and list its notable characteristic. 8

(b) The hot junction of an iron- constanton thermocouple is connected to a potentiometer whose terminals are at 25°C. The potentiometer reads 3.6 mV. Work-out the temperature at the hot function of the thermocouple. It may be assumed that the couple conforms to the following values which are based reference junction at 0 ºC.

2018

Temperature	-20	10	40	70	100	130
Voltage mV	-0.8	0.5	1.5	3.41	4.91	6.42

8

5. (a) List Various flow measuring devices. Sketch and explain the working of Magnetic flow meter. 8

2018

(b) State the objective of flow-visualisation. Explain some of the methods commonly adopted for flow visualisation in low speed flow. 8

2019

Cont.

6 (a) Describe the principle of operation variable of linear variable
2019 differential transformer (LVDT). Sketch the typical input and output graph. 8

(b) Explain the function of a dummy gauge in a strain gauge load cell. Draw the circuit arrangement of a bridge employing two strain gauges for the measurement of strain of a cantilever beam so that no error is caused because of change in gauge resistance due to environmental temperature change. 8

7 Write short notes on any four -

2019 (i) Use of strain gauge on rotating shaft 2018

2019 (ii) CRO

PYQ 2019 (iii) Bridgeman Gauge

(iv) RTD & thermister → 2018, 19

(v) pneumatic load cell

PYQ 2019 (vi) Reed Vibrometer

4x4=16

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Roll No. 22UMEC1134

BE II (ME)

3 **RAC**

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

ME 251 A : Refrigeration & Air Conditioning (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

Use of Psychometric charts and Refrigeration table and P-h charts and steam table are permitted.

1. (a) Derive the expression for COP of a reversed Carnot refrigeration cycle. What are the limitations of the cycle? P40-18

8

(b). Describe boot-strap cycle of air refrigration system, with a schematic diagram and show the cycle on T-S diagram. P40-19

8

PYQ-18

2.(a) Explain the effect of superheating in compressor and under cooling in vapour compression system. Show these on T-S and P-h diagram. 8

(b) A refrigeration machine using R-12 as refrigrant operates between the pressure 2.5 bar and 9 bar. The compression is isentropic and there is no undercooling in the condenser.

The vapour is in dry saturated condition at the beginning of the compression. Estimate the theoretical coefficient of performance. If the actual COP is 0.65 of the theoretical value, calculate the net cooling produced per hour. The refrigerant flow is 5 kg per minute.

Properties of refrigerant are:

Pressure (bar)	Saturation temp' °C	Enthalpy (KJ/kg)		Entropy of Saturated Vapour (KJ/KgK)
		Liquid	Vapour	
9.0	36	70.55	201.8	0.6836
2.5	-7	29.62	184.5	0.7001

8

3. (a). Derive an expression for the COP of an ideal vapour absorption system is terms of temperature T_G at which heat is supplied to the generator, the temp' T_E at which

(Contd.)

heat is absorbed in the evaporator and the temp^r T_c at which heat is discharged from the condenser and absorber. 8

(b) Draw a neat diagram of lithium bromide water absorption system and explain its working. 8

4. (a) What do you mean by ODP and GWP related to refrigerants? Discuss the physical and thermodynamic properties of the ideal refrigerant. 8

(b) A simple air cooled system is used for an aeroplane having a load of 10 tonnes. The atmospheric pressure and temp^r are 0.9 bar and 10°C respectively. The pressure increases to 1.013 bar due to ramming. The temp^r of air is reduced by 50°C in the heat exchanger. The pressure in the cabin is 1.01 bar and the temp^r of air leaving the cabin is 25°C. Determine :

(i) Power required to take the load of cooling in the cabin

(ii) COP of the system

Assume that all the expansion and compressions are isentropic. The pressure of the compressed air is 3.5 bar.

5. (a) Establish the following expressions for air vapour

P40-15 mixture:

Specific humidity $w = 0.622 \times \dfrac{P_v}{P_b - P_v}$

P_v = Partial Pr. of water vapour

P_b = Barometric pressure 4

(b) For a sample of air having 22°C DBT, relative humidity 30% at barometric pressure of 760 mm of Hg, Calculate

(i) Vapour Pressure

(ii) Humidity ratio

(iii) Enthalpy 6

(c) At a certain locality, the dry bulb temp' of air is 30°C and the relative humidity is 40%. Determine the specific humidity and dew point and wet bulb temp' of air. If the air is cooled in an air washer using recirculated spray water and having a humidifying effeciency of 0.9. What are dry bulb temp' and dew point temp' of air leaving the air washer? 6

6. (a) Explain with a neat sketch the working of Year - Round

P40-19 Air Conditiong System. 8

(b) What is "effective temperature"? What factors affect

P40-19, 15 effective temp'? 8

7. Write short notes on any four 4

(i) Dry air rated temp' (DART)

(ii) Automatic expansion value

(iii) Electrolux refrigerator

(iv) Comfort charts PYQ-19

(v) Chemical dehumidification

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Roll No. 22UMEC1134...

BE II (ME)

3

F&WE

SECOND B.E. EXAMINATION, 2023
MECHANICAL ENGINEERING
(IV SEMESTER) (CBCS)
ME 252 A : Foundry & Welding Engineering(M)

Time – Three Hours

Maximum Marks – 80

Note: (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Consider a foundry for steel castings. What types of pattern would you suggest for small scale, medium scale and large scale production? Also Suggest various Pattern allowances for them. 10

(b) What are the various test methods for moulding sand to check the quality of sand? Describe any four methods. 6

2. (a) Explain in detail how investment castings are produced with suitable diagrams. 10

(b) Compare hot chamber with cold chamber die casting mentioning various advantages and disadvantages. 6

3. (a) What are the various casting defects? How can we overcome or reduce them? 6

(b) Discuss advantages, limitations and applications

(i) Carbon dioxide moulding
(ii) Shell moulding
(iii) Centrifugal casting
(iv) Continuous casting 10

4. (a) How metal transfer in welding occurs? Explain metal transfer methods with suitable diagrams. 10

(b) Compare TIG and MIG welding also discuss their applications. 6

5. (a) What are various flame types in oxy acetylene gas welding show with proper diagram and various zones? 8

(b) Explain various welding joints and welding positions with suitable diagrams. 8

6. (a) Attempt any four, write short notes:- 4x4

(i) Soldering and Brazing
(ii) Gate, runner, riser
(iii) Cupola furnace
(iv) Friction welding
(v) Thermal welding
(vi) AC & DC welding and their applications

7. (a) What are various resistance welding process. Discuss briefly with diagram. 10

(b) Non destructive testing used in both casting and welding. Explain any three methods. 6

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Roll No. 221/MEC/1134

BE II (ME)

Kine. & Dyna. of Mach.

SECOND B.E EXAMINATION - 2023

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

ME - 253 : Kinematics & Dynamics of Machines

(M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Q. No. 1 complete solution (including calculation) is to be given in the drawing sheet.

(iii) All questions carry equal marks.

(iv) Assume suitable data, if not given and mention it clearly.

(v) Draw neat sketches, wherever necessary.

3520

1. A flat-faced mushroom follower is operated by a uniformly rotating cam. The follower is raised through a distance of 35mm in 120° rotation of the cam, remains at rest for the next 30° and is lowered during further 90° rotation of the cam. The raising of the follower takes place with cycloidal motion and the lowering with uniform acceleration and deceleration. However, the uniform acceleration is 2/3of the uniform deceleration. The least radius of the cam is 25 mm which rotates at 450 rpm. Draw the cam profile and determine the values of the maximum velocity and maximum acceleration during rising, and maximum velocity and uniform acceleration and deceleration during lowering of the follower.

16

2. (a) In a Hartnell governor, the radius of rotation of the balls is 50 mm at the minimum speed of 280 rpm. The length of the ball arm is 130 mm and the sleeve arm is 80 mm. The mass of each ball is 3 kg and the sleeve is 5 kg. The stiffness of the spring is 25 N/mm. Determine the (i) speed when the sleeve is lifted by 55 mm (ii) initial compression of the spring (iii) governor effort (iv) power.

8

(b) What is the controlling force of a governor? How are the controlling force curves drawn? How do they indicate the stability or instability of a governor? Indicate the shape of such a curve for an isochronous governor. 8

3. (a) A band and block brake having 12 blocks, each of which subtends an angle of 18° at the centre, is applied to a rotating drum with a diameter of 600 mm. The blocks are 75 mm thick. The drum and the flywheel mounted on the same shaft have mass of 1900 kg and have a combined radius of gyration of 600 mm. The two ends of the band are attached to pins on the opposite sides of the brake fulcrum at distances of 42 mm and 158 mm from it. If a force of 300 N is applied on the lever at a distance of 950 mm from the fulcrum, find the (i) maximum braking torque (ii) angular retardation of the drum (iii) time taken by the system to the stationary from the rated speed of 350 rpm.

Take coefficient of friction between the blocks and the drum as 0.32. 8

(b) Describe with the help of a neat sketch the principles of operation of an internal expanding shoe. Derive the expression for the braking torque. 8

4. (a) Describe the construction and operation of a prony brake or rope brake absorption dynamometer with neat sketch. 8

(b) A torsion dynamometer is fitted on a turbine shaft to measure the angle of twist. It is noticed that the shaft twist 2.5° in a length of 5 meters at 750 rpm. The shaft is solid and has a diameter of 250 mm. If the modulus of rigidity for the shaft is 82 GPa, find the power transmitted by the machine. 8

5. (a) A four - wheeled trolley car has a total mass of 3500 kg. Each axle with its two wheels and gears has a total moment of inertia of 38 kg.m^2. Each wheel is of 450 mm radius. The centre distance between two wheels on an axle is 1.6m. Each axle is driven by a motor with a speed ratio of 1:3. Each motor along with its gear has a moment of inertia of 19 kg.m^2 and rotates in the opposite direction to that the axle. The centre of mass of the car is 1.2 m above the rails. Calculate the limiting speed of the car when it has to travel around a curve of 250 m radius without the wheels leaving the rails. 8

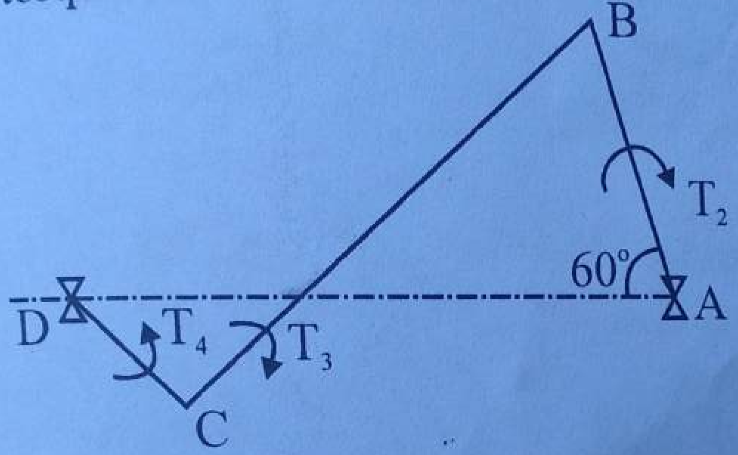
(b) Explain the gyroscopic effect on ships. Also explain the axis of spin, precession, and gyroscopic couple with neat sketches.

J- 1153

(Contd.)

6/2/0

6 (a) In a four-link mechanism shown in Figure 1, torque T3 and T4 have magnitude of 35 N.m and 25 N.m, respectively. The link lengths are AD = 850 mm, AB = 350 mm, BC = 750 mm and CD = 450 mm. For the static equilibrium of the mechanism, determine the required input torque T2.



4 (b) Define and explain the superposition theorem as applicable to a system of forces acting on a mechanism. 8

4x4 = 16

7. Write short notes (any four).

(a) Describe the function of a simple Watt governor. What are its limitations?

(b) Explain the tractive resistance in wheeled vehicle.

(c) What is the principle of virtual work? Explain.

(d) Explain the gyroscopic effect on four-wheeled vehicles.

(e) Discuss various types of cams and draw neat sketches.

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Roll No. 22UMEC1134

BE II (ME)

3 MD - I

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

ME 254 A : Machine Design - I (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

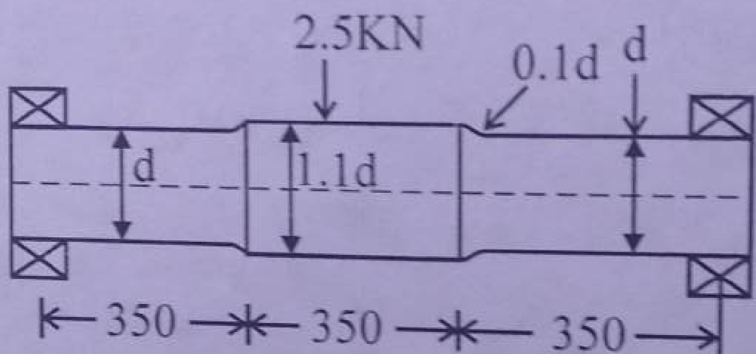
(iii) All questions carry equal marks.

1. (a) Define machine design and morphological analysis used in Designing Process. 8

(b) Describe the designers responsibility and various aspects of design used in Machine design. 8

2.(a) Why we required different theories of failure in machine design? Explain different theories of failure. 8

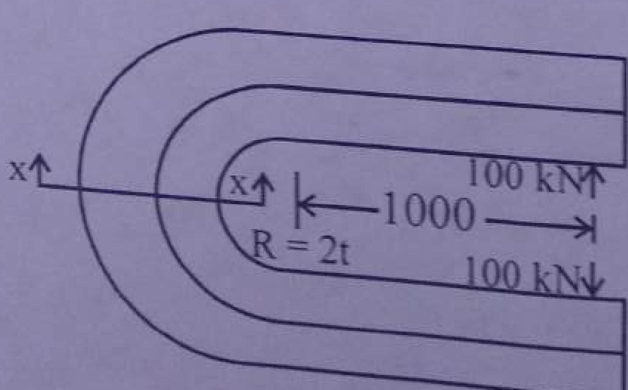
(b) A non - rotating shaft supporting a load of 2.5 kN as shown in Fig. The shaft is made of brittle material, with an ultimate tensile strength of 300 N/mm^2. The factor of safety is 3. Determine the dimensions of the shaft. 8



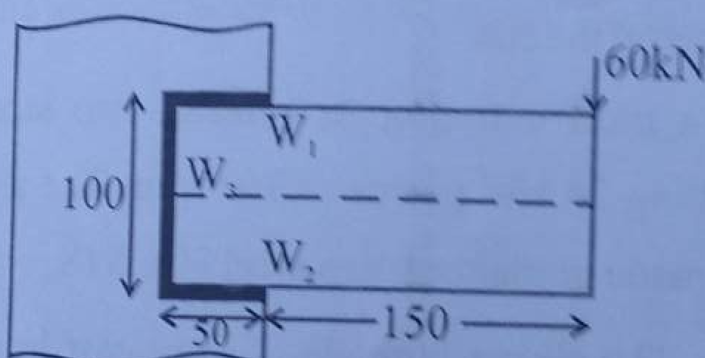
3. The C - frame of a 100 kN capacity press is shown in Fig. The material of the frame is grey cast iron FG 200 and the factor of safety is 3. Determine the dimensions of the frame.

4.8

16



4. 8.10 A welded connection as shown in fig. is subjected to an eccentric force of 60 kN in the plane of the welds. Determine the size of the welds, if the permissible shear stress for the weld is 100 N/mm^2. Assume static conditions. 16



5. Q.9 A safety valve, 50 mm in diameter is to blow off at a pressure of 1.5 NPa. It is held on its seat by means of a helical compression spring with an initial compression of 25 mm. The maximum lift of the valve is 10 mm. The spring index can be taken as 6. The spring is made of patented and cold-drawn steel wire with ultimate tensile strength of 1500 N/mm^2 and modulus of rigidity of 81370 N/mm^2. The permissible shear stress for the spring wire should be taken as 30% of the ultimate tensile strength. Design the spring and calculate.

(i) Wire diameter

(ii) Mean coil diameter

(iii) No. of active turns

(iv) Total no. of turns

(v) Solid length

(vi) Free length

(vii) Pitch at the coil 16

6. Design a muff coupling to connect two steel shafts transmitting 25 kW power at 360 rpm. The shafts and key are made of plain carbon steel 30 C8 ($S_{yt} = S_{yc} = 400$ N/mm^2). The sleeve is made of grey cast iron FG200 ($S_{ut} = 200$ N/mm^2). The factor of safety for the shaft and key is 4. For the sleeve the factor of safety is 6 based on ultimate strength. 16

7. A right angled bell- crank lever is to be designed to raise a load of 5 kN at the short arm end. The lengths of short and long arm are 100 and 450 mm respectively. The lever and the pins are made of steel 30 C8 ($S_{yt} = 400$ N/mm^2) and the factor of safety is 5. The permissible bearing pressure on the pin is 10 N/mm^2. The lever has a rectangular corss-section and the ratio of width to thickness is 3:1. The length to diameter ratio of the fulcrum pin is 1.25:1. Calculate.

4,16

(i) The diameter and the length of the fulcrum pin.

(ii) The shear stress in the pin.

(iii) The dimensions of the boss of the lever at the fulcrum.

(iv) The dimensions fo the cross-section of the lever.

Assume that the arm of the bending moment on the lever extends up to the axis of the fulcurm. 16

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Roll No. 22UMEC1134

BE II (ME)

3

FM - I

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(IV SEMESTER CBCS)

ME 255 A : Fluid Mechanics - I (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

(iv) Whatever data are missing, assume and mention them clearly.

1.(a) (i) Explain the effect of temperature on viscosity of liquids and gases. 4

(ii) Define Newtonian and Bingham fluid graphically. Give at least two examples for each of them. 4

(Contd.)

(b) Two horizontal flat plates are placed 0.15 mm apart and the space between them is filled with an oil of viscosity 1 poise. The upper plate of area 1.5 m^2 is required to move with a speed of 0.5 m/s relative to the lower plate. Calculate the necessary force and power required to maintain this speed. 8

2.(a) Derive an expression for hydrostatic force acting on a submerged plane surface inclined with respect to free surface of liquid. 8

(b) Determine the difference of pressure between pipes A and B when connected to an inverted U-tube differential manometer containing oil of specific gravity 0.8 as the manometric liquid. The pipe A conveys water and a fluid of specific gravity 0.9 flows through the pipe B. The position of manometric liquid in the manometer limbs is indicated in figure - 1

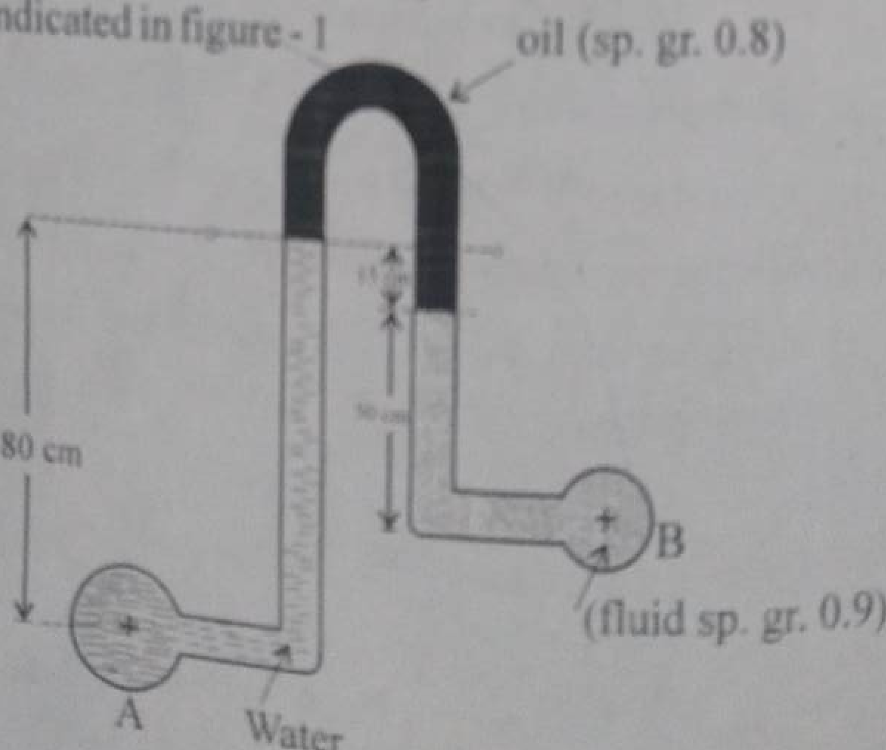

Figure - 1

If P_B (gauge) $= 5\times10^4$ N/m^2 and the barometer reading is 730 mm of mercury, find the pressure in pipe A in meters of water absolute. 8

3.(a) What do you understand by metacenter of a floating body? Define metacentric height. What are the conditions of equilibrium of floating bodies? 8

(b) A wooden block of width 15cm x depth 30 cm and length 150 cm has a mass of 54 kg. Can the block float with 30 cm side vertical in water? Comment on the stability of the block. 8

4.(a) Derive continuity equation in three dimensions for Cartesian coordinates for general flow. Simplify it for various cases of flow. 8

(b) The velocity component in a two dimensional flow field for an incompressible fluid are expressed as

$$u = \frac{y^3}{3} + 2x - x^2 y$$

$$v = xy^2 - 2y - \frac{x^3}{3}$$

(i) Show that these functions represent a possible flow.

(ii) Show that these functions represent an irrotational flow.

(iii) Obtain an expression for stream function ψ.

(iv) Obtain an expression for velocity potential ϕ. 8

J-1212

5.(a) Derive Euler's equation of motion along a stream line for an ideal fluid stating clearly all the assumptions. 8

(b) Prove that discharge through a right angled triangular V-notch is given by

$$\theta = \frac{8}{15} C_d \sqrt{2g}\, H^{5/2}$$

Where symbols have their usual meanings. 8

6.(a) What is the principle of Venturimeter? Explain. Derive an expression for discharge through a venturimeter. Why actual discharge is less than theoretical discharge?

8

(b) Consider force F on propeller of an aircraft. It depends on forward speed of aircraft U, density of air ρ, viscosity of air μ, diameter of propeller D and speed of rotation of propeller N. Show by dimensional analysis using Buckingham's π Theorem:

$$\frac{F}{\rho U^2 D^2} = f\left[\frac{UD\rho}{\mu}, \frac{ND}{U}\right]$$

8

7. Write short notes on the following : 4x4=16

(a) Stream line, path line and streak line.

(b) Measurement of static, stagnation and dynamic pressure.

(c) Froude Number and Weber number & their applications.

(d) Vorticity and Circulation.

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Roll No. 22UMEC1134

BE III (ME)

3 **FM-II**

THIRD B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(V SEMESTER CBCS)

ME 301 A: FLUID MECHANICS II

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Derive the formula for calculating loss of head due to sudden expansion. 8

(b) A pipe of 5 cm diameter is 5 m long and carries a discharge of 0.005 m^3/s. What loss of head and corresponding power would be saved if the central 2m length of pipe was replaced by 7.5 cm diameter of pipe, the change in section being sudden. Take friction factor as 0.04 for pipe of both diameters and fluid is water. 8

2. (a) Find an expression for power transmission through pipes. What is the condition of maximum transmission of power and corresponding efficiency of transmission? 8

(b) Two pipes of the same material and of equal lengths are available for connection to an overhead tank which can supply $0.085\ m^3/s$ of water. The diameter of the pipes are 30 cm and 15 cm, respectively. Determine the ratio of head loss when the pipes are connected in series to the loss of head when they are connected in parallel. Neglect minor losses. 8

3. (a) Derive an expression for velocity distribution for viscous flow through a circular pipe. Sketch the velocity distribution and shear stress distribution across a section of the pipe. 8

(b) Write short notes on the following : 2x4=8

(i) Conette flow

(ii) Prandtl mixing length hypothesis.

4. (a) For the velocity distribution in laminar boundary layer as :

$$\frac{u}{U} = \frac{3}{2}\left(\frac{y}{\delta}\right) - \frac{1}{2}\left(\frac{y}{\delta}\right)^3$$

Find the momentum thickness in terms of δ. Using Von-Karman integral momentum equation prove that

$$\delta/x = \frac{4.64}{\sqrt{Re_x}}$$

where symbols have their usual meanings. 10

(b) What are the causes of separation in boundary layer? How the separation can be controlled? 6

5. (a) A flat plate is placed parallel to a uniform stream of air. Assuming boundary layer to be turbulent over the entire plate. Determine the ratio of drag force on leading half and rear half of the plate. Assume local skin friction coefficient,

$$Cf_x = \frac{0.0576}{Re_x^{0.2}}$$ 8

(b) A Smooth flat plate 2m wide and 2.5 m long is towed in oil (sp.gravity 0.8) at a velocity of 1.5 m/s. Find boundary layer thickness and shear stress at (i) Centre of plate , (ii) at trailing edge of the plate. Find power required to tow the plate. Kinematic viscosity of oil = $10^{-4} m^2/s$. 8

6. (a) Define with schematic diagram the following nomenclature for an unsymmetrical aerofoil.

   (i) Chord line

   (ii) Camber line

   (iii) Angle of attack

   (iv) Span

   (v) Aspect ratio 8

   (b) Explain the phenomenon of Karman vortex street for the flow around a bluff body. What do you mean by Strouhal number. 8

7. Write short notes on the following.

   (a) Skin friction drag and pressure drag.

   (b) Hydraulic gradient line and total energy line.

   (c) Flow around a sphere and Stoke's law.

   (d) Darey's Weisbach Equation. 4x4 = 16

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BE III (ME)

3

PMT

THIRD B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(V SEMESTER CBCS)
ME-302 A: PRODUCTION MACHINE TOOLS

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Explain constructional details and operations on Turret lathe. 8

PYQ-23

(b) What is quick return mechanism? Explain with neat sketch. Also write its applications. 8

2. (a) Explain constructional details of universal milling machine with its applications. 8

PYQ-23

(b) What is Indexing? Explain different indexing methods. 8

PYQ-22

3. (a) Differentiate between Jigs and Fixtures. Explain principles of location and clamping. 8

PYQ-23

(b) What is need of dressing and truing of grinding wheels? Explain different methods of dressing & truing of grinding wheels. 8

4. (a) What do you mean by Automats? Explain their operation planning and tool layout. 8

(b) What is bulk deformation? Explain different type of dies. 8

5. (a) What is the difference between NC and CNC System? Explain various components of CNC System. 8

(b) Explain position and velocity feedback devices. 8

6. (a) What is adaptive control? Explain its importance in machining operation. 8

(b) Explain construction details of radial drilling machine. 8

7. Write short note on any of two of the following questions. 2X8=16

(a) Lapping and Honning

(b) Machine control unit

(c) Hydraulic tracer controlled machine tools

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BE III (ME)

3

IO&M

THIRD B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(V SEMESTER CBCS)
ME-303 A: INDUSTRIAL ORGANIZATION & MANAGEMENT

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Compare Private and Public Limited Company. 8

   (b) Describe formation and working of Public Sector undertaking. 8

2. (a) Describe functions and elements of management. 8

   (b) Explain structure of line and staff organization with suitable charts. 8

3. (a) List 14 principles of management given by Fayol. Describe any five out of them. 8

(b) Write difference between following: 2X4=8

(i) Authority and Responsibility

(ii) Long term and short term planning

(iii) Sole Properietorship and Partnership

(iv) Formal and Informal Organization

4. (a) Describe leadership functions and types of leaders. 8

(b) Explain different distribution channels. 8

5. (a) Briefly describe different investment appraisal criteria. 8

(b) Describe balance sheet and profit and loss account. 8

6. (a) Explain job specification and job qualification. 8

(b) Describe recruitment and selection. 8

7. Write short notes (any four) 4X4=16

(i) Conflict

(ii) Advertising

(iii) Ratio analysis

(iv) Decentralization

(v) Organization chart

(vi) Training methods

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BE III (ME)

3

Dyn. of Mach. (M)

THIRD B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(V SEMESTER CBCS)
ME-304 A: DYNAMICS OF MACHINES - (M)

Time - Three Hours

Maximum Marks - 80

Note:-
(i) Attempt any **Five** questions.
(ii) All questions carry equal marks.
(iii) Assume suitable data. if not given and mention it clearly.
(iv) Draw neat sketches, wherever necessary.

1. (a) A riveting machine is driven by a motor of 5 KW. The actual time to complete one riveting operation is 1.8 seconds and it absorbs 15 KN.m of energy. The moving parts including the flywheel are equivalent to 180 kg at 350 mm radius. Determine the speed of the flywheel immediately after riveting it is 540 rpm before riveting. Also, find the number of rivets closed per minute. 8

P40 Value Change

(b) Explain procedure to construct turning moment diagram of a four stroke IC engine with neat sketches. 8

P40-23

2. (a) From the first principle, determine expression for displacement, velocity and acceleration of piston in an IC engine. 8

PYQ-2023

(b) In a single - slider crank mechanism, the force acting on slider is 4000 N (inward direction) and coefficient of friction between all the links is 0.3. Calculate the driving torque if the pin diameters at joints O, A and B are 80 mm, 80 mm and 40 mm respectively. The dimensions of links are: OA = 300 mm, AB = 1050 mm and $\angle BOA = 60°$. 8

3. (a) Describe the function of a pivoted-cradle balancing machine with the help of a neat sketch. Show that it is possible to make only four test runs to obtain the balance masses in such a machine. 8

(b) The following data refer to a two-cylinder uncoupled locomotive: Rotating mass per cylinder is 300 kg; reciprocating mass per cylinder is 350 kg; distance between wheels is 1500 mm; distance between cylinder centres is 650 mm; diameter of treads of driving wheels is 180 mm; crank radius is 350 mm; radius of centre of balance mass 650 mm; locomotive speed is 50 km/hr; angle between cylinder cranks is 90°; dead load on each wheel is 3.5 tonne. Determine the: (i) balancing mass required in the planes of driving

wheels if whole of the revolving and two-third of the reciprocating mass are to be balanced. (ii) swaying couple (iii) variation in the tractive force (iv) maximum and minimum pressure on the rails; (v) maximum speed of locomotive without lifting the wheels from the rails. 8

4. (a) What is an epicyclic and reverted gear train? Explain with suitable sketches. 8

(b) In a sun and planet gear train, the sun gear wheel having 85 teeth is fixed to the frame. Determine the number of teeth on the planet and annulus wheels if the annulus rotates 160 times and the arm rotates 120 times, both in the same direction. 8

5. (a) A shaft carries four masses in P, Q, R and S which are placed in parallel planes perpendicular to the longitudinal axis. The unbalanced masses at planes Q and R are 3.2 kg and 2.1 kg respectively and both are assumed to be concentrated at a radius of 28 mm while the masses in planes P and S are both at radius of 50 mm. The angle between the planes Q and R is 110° and that between Q and P is 210°, both angles being measured in counter - clockwise direction from the plane Q. The planes containing P and Q are 280 mm apart those containing Q and R are 550 mm. If the shaft is to be completely balanced, determine: (i) masses at the planes P and S: (ii) the distance between the planes R and S; (iii) the angular position of the mass S. 8

(b) Explain the terms primary balancing and secondary balancing used for balancing of reciprocating masses in reciprocating engine mechanism. 8

PYQ-23

6. (a) Show that for a rack and pinion arrangement with 20° pressure angle, the minimum number of teeth required on pinion to avoid interference is 18. 8

(b) A worm gear set transmits power between two shafts which are at right angle to each other. The gear ratio is 6 and the worm has 4 teeth of normal pitch 20 mm. The pitch diameter of the worm is 50 mm. Calculate the tooth angles of the worm and worm gear and the distance between the shafts. If the efficiency of the drive is 85 percent, the worm being the driver, find the coefficient of friction between the matching tooth surfaces. 8

7. Write short notes (any four). 4X4

(a) What do you mean by free body diagram? Explain with suitable example.

(b) Explain clearly the terms static and dynamic balancing.

PYQ (c) Explain law of gearing.

(d) What is field balancing of rotors? Explain the procedure.

PYQ (e) 23 Explain the procedure adopted for designing the helical gears.

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BE III (ME)

3

MD - II

THIRD B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(V SEMESTER CBCS)
ME-305 A: MACHINE DESIGN - II

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

(iv) Assume suitable data

1. Design a shaft to transmit power from motor to a lathe head stock through a pulley by means of a belt drive. The pulley weighs 200N and is located at 300mm from the centre of bearing. The diameter of pulley is 200 mm and maximum power transmitted is 1kW at 120 RPM. The angle of lap of belt is 175° and coefficient of friction between belt and pulley is 0.3. The shock and fatigue factors for bending and twisting are 1.5 and 2.0 respectively. The allowable shear stress in the shaft is 40 MPa. 16

2\. (a) Compare belt drive, chain drive and gear drive. 6

(b) Design completely a belt drive to drive a winch from an electric motor of 11 KW. Power. Speed of motor shaft is 750 rpm. Speed ratio is 4. belt position is horizontal and there is considerable variation of load. 10

3\. Two helical gears are used in speed reducer that is to be driven by an IC engine. The rated power is 75 KW at a pinion speed of 1200rpm. The speed ratio is 3. Design the drive. 16

4\. Design a worm gear drive to transmit 11.25 KW from a shaft rotating 1500 rpm to another shaft at 75 rpm. 16

5\. Design screw jack for lifting height of 100mm and load of 400 KN. 16

6\. (a) A multi disk clutch consists of five steel plates and four bronze plates. The inner and outer diameters pf the friction disks are 75 and 150 mm respectively. The coff. of friction is 0.1 and the intensity of pressure on friction lining is limited to 0.3 N/mm^2. Assuming uniform wear theory calculate:

(i) The required force to engage the clutch 2650.71N

(ii) Power transmitted capacity at 750 rpm. 10 Mt = 119282.3 Nmm

→ 9.3

(b) Compare disk brake and drum brake. 6

7. Attempt any two of the following: 8X2

(a) Define power screw and their applications.

(b) Prove that a hollow shaft has greater strength and stiffness than a solid shaft of equal weight.

(c) Explain significance of uniform pressure theory and uniform wear theory for clutches.

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Roll No. 2212MEC1134

BE III (EE)

1

Open Elect.-II

THIRD B.E. EXAMINATION - 2024
ELECTRICAL ENGINEERING
(V SEMESTER CBCS)
OPEN ELECTIVE-II:
EE-342 A: ENERGY CONSERVATION

Time - Three Hours
Maximum Marks - 100

Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Discuss the objectives of energy management. 10
   (b) Describe former scope for energy conservation and its benefits. 10
2. (a) Explain the Energy Conservation Act and its features. 10
   (b) List air pollution control equipment and explain any one in detail. 10
3. (a) State the benefits of power factor improvement and explain any one technique to improve power factor in an electrical system. 10

(b) Explain the effect of harmonics on motors and remedies leading to energy conservation. 10

4. (a) Explain need and type of energy audit. 10

(b) Prepare a list of energy audit instruments and explain any one. 10

5. (a) Discuss the measures to optimise transmission and distribution losses. 10

(b) Explain types of tariff and restructuring of electrical tariff techniques. 10

6. (a) Explain automatic power factor controller used in electrical system. 10

(b) Describe about energy efficient transformers and lighting controls. 10

7. Write short note on any two :

(a) Demand - supply gap 10

(b) Cogeneration 10

(c) Soft starters with energy saver 10

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BE III (ME) **Roll No.** 22U/MEC/134

3 **Steam Power Engg.**

THIRD B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(VI SEMESTER)

ME 351 A : STEAM POWER ENGINEERING (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Prove that for maximum mass of hot gases discharged through the chimney,

$$\frac{T_2}{T_1} = \frac{2(m+1)}{m}$$

where T_1 = Absolute tempr. of Cold air

T_2 = Absolute tempr. of Hot air, and

M = Mass of air supplied (Kg/Kg of fuel) 6

(b) A chimney of 60 meter high and the temperature of atmospheric air is 27°C. If 15 kg of air/kg of fuel is used, Find for maximum discharge of hot gases:

(i) The temperature of hot gases, and

(ii) The draught pressure in mm of water

4

(c) In a boiler, the following observations were made :

X

steam pressure	= 10 bar
steam condensed	= 540 kg/hour
fuel used	= 65 kg/hr
Moisture in fuel	= 2% by mass
Mass of dry flue gases	= 9 kg/kg of fuel
lower calorific value of fuel	= 32000 kJ/kg
Tempr. of the flue gases	= 325°C
Tempr. of boiler house	= 28°C
Feed water temperature	= 50°C
Mean specific heat of flue gases	= 1 kJ/kg k
Steam Dryness fraction	= 0.95

Draw up a heat Balance sheet for the boiler. 6

2. (a) Derive an expression for maximum discharge through convergent divergent nozzle for steam. 10

(b) Steam at a pressure of 10 bar and 210°C is supplied to a convergent divergent nozzle with a throat area of 1500 mm^2. The exit is below critical pressure. Find the coefficient of discharge, if the flow is 7200 kg of steam per hour. 6

3. (a) Dry saturated steam enters a nozzle at a pressure of 10
X bar and with on initial velocity of 90 m/s. The outlet pressure is 6 bar and the outlet velocity is 435 m/s. The

heat loss from the nozzle is 9 kJ/kg of steam flow,

Calculate the dryness fraction and the area at the exit, if the area at the inlet is 1256 mm^2. 8

(b) Dry saturated steam at a pressure of 8 bar enters a convergent - divergent nozzle and leaves at a pressure of 1.5 bar. If the flow is isentropic (expansion index = 1.135), Find the ratio of cross sectional area at exit and throat for maximum discharge. 8

4. (a) Discuss the method of velocity compounding of an impulse turbine for achieving rotor speed reduction. 6

(b) At a stage of a reaction turbine, the rotor diameter is 1.4m and speed ratio 0.7. If the blade outlet angle is 20° and the rotor speed is 3000 rpm, find the blade inlet angle and diagram efficiency.

Also find the percentage increase in diagram efficiency and rotor speed, if the turbine is designed to run at the best theoretical speed. 10

5. (a) Discuss the saving in Heat Rate from regenerative heating. 4

(b) Dry and saturated steam enters a steam turbine at 40 bar and exhausts at 0.07 bar. It is planned to use a regenerative feed heating system employing three heaters.

(i) Design suitable extraction points and estimate the mass of steam taken by the heater per kg of feed.

(ii) Find efficiency of the regenerative cycle. 12

(Contd.)

6. (a) Define the term "Degree of reaction" as applied to a reaction turbine. Show that for a Parson's reaction turbine, the degree of reaction is 50 percent. 6

(b) At a certain pair in a reaction turbine, the steam leaves the fixed blade at a pressure of 3 bar with a dryness fraction 0.98 and a velocity 130 m/s. The blades are 20 mm high and discharge angle for both the rings is 20°. The ratio of axial velocity of flow to the blade velocity is 0.7 at inlet and 0.76 at exit from the moving blade. If the turbine uses 4 kg of steam per second with 5% tip leakage, find the mean blade diameter and the power developed in the ring. 10

7. (a) What do you mean by dry cooling towers? With the help of sketches, discuss the various types of dry-cooling tower. 6

(b) The following observations were made during a test on a steam condenser.

Barometer reading	=	765 mm of Hg
Condenser vacuum	=	710 mm of Hg
Mean condenser temperature	=	35°C
Condensate temperature	=	28°C
Condensate collected per hour	=	2 tonnes
Quantity of cooling water per hour	=	60 tonnes
Tempr. of cooling water at inlet	=	10°C
Tempr. of cooling water at outlet	=	25°C

Find

(i) Vacuum corrected to the standard barometer reading.

(ii) Vacuum efficiency of the condenser.

(iii) Under cooling of the condensate.

(iv) Condenser efficiency.

(v) Quality of steam entering the condenser,

(vi) Mass of air per m^3 of condenser volume, and

(vii) Mass of air per kg of uncondensed steam. 10

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Roll No. 22UMEC1134

BE III (ME)

3

MC&M

THIRD B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(VI SEMESTER)
ME 352 A: METAL CUTTING & METROLOGY
(M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Explain theory of chip formation and chip flow. 8

(b) During turning a carbon steel rod with a carbide tool of geometry 7, _, 6, 6, 8, 30, 1 (inch) at speed of 420 rpm, the following observations were made:

Uncut chip thickness, t = 2 mm

Width of chip, b = 2.5 mm

Cutting force, $F_c = 1177$ N

Thrust force, $F_t = 560$ N

For the above machining conditions, determine : $\lambda = 60$, $\alpha = 13.71$

(i) Side rake angel, α_s → 18.36, F = 823 N, N = 1010.74 N

(ii)C-efficient of friction, μ = 0.814

ϕ = 39.152

(iii)Dynamic shear strength of the work material. F_s = 1178, 8

τ = 148.76

2. (a) Explain tool wear with location and mechanism. 8

(b) What is Taylor's equation for tool life? In a certain machining operation with a cutting speed of 50 m/min, tool life of 45 minutes was observed. When the cutting speed was increased to 100 m/min, the tool life decreased to 10 minutes. Estimate the cutting speed for maximum productivity, if tool change time is 2 minutes. 8

n = 0.30, V_3 = 162.9 m/min, C = 200, 156

3. (a) What are comparators? Explain optical comparator. 8

(b) Calculate the dimensions of plug and ring gauges to control the production of 50 mm shaft and hole pair of H_7d_8 as per IS specifications. The following assumptions may be made : 50 mm lies in diameter step of 30 - 50 mm. Upper deviation for shaft 'd' is $-16D^{0.44}$ → 79.94 and lower deviation for hole 'H' is zero. Tolerance unit 'i' in Microns is $= 0.45\,(D)^{1/3} + 0.001D$ and IT7=16i and, 0.024 0.038 IT8 = 25i. Also indicate these limits and tolerances on a diagram. 8

D = 38.72 mm, i = 1.542

Shaft max 50.117, min 50.079; hole max 50.024 mm, min 50 mm

4. (a) What is optical Interferometry? Explain. 8

(b) In the analyze phase, a project team collected 10 sets of NUT diameter samples (mm) with a subgroup size of 10 from a manufacturing unit. Prepare X-bar and R chart for the given sample of product and comment on the results. For $n = 10; A_2 = 0.31, D_3 = 0.22$ and $D_4 = 1.78$.

S.No.	Measured Values (mm)									
	1	2	3	4	5	6	7	8	9	10
1	9.24	9.33	9.33	9.32	9.33	9.25	9.32	9.32	9.33	9.33
2	9.34	9.20	9.33	9.22	9.24	9.23	9.33	9.33	9.33	9.32
3	9.32	9.27	9.19	9.23	9.20	9.33	9.34	9.18	9.32	9.33
4	9.33	9.32	9.32	9.34	9.26	9.33	9.33	9.31	9.24	9.25
5	9.32	9.35	9.34	9.24	9.34	9.37	9.23	9.23	9.33	9.32
6	9.33	9.33	9.24	9.23	9.32	9.33	9.33	9.34	9.27	9.35
7	9.34	9.32	9.32	9.32	9.33	9.33	9.26	9.24	9.32	9.32
8	9.32	9.34	9.32	9.32	9.24	9.33	9.32	9.32	9.24	9.32
9	9.19	9.17	9.33	9.33	9.26	9.32	9.33	9.33	9.33	9.32
10	9.26	9.35	9.33	9.31	9.34	9.28	9.34	9.33	9.23	9.34

5. (a) Explain the key measurements and specifications for a gear, including the outside diameter, tooth thickness, depth of gear tooth, pitch circle diameter, module, circular pitch, addendum, dedendum and chord width. 8

(b) What do you mean by acceptance sampling? Explain single and multiple sampling plans. 8

6. (a) Explain different cutting parameters and the their effects on forces, power and surface finish. 8

(b) A single point cutting tool with 12° rake angle is used to

(Contd.)

machine a steel work piece. Depth of cut i.e. uncut chip thickness is 0.81 mm and chip thickness is 1.8mm. Find the value of shear angle. Also find the cutting shear strain for above machining conditions. 8

7. Attempt any two questions of the following:

(a) Explain Orthogonal and oblique cutting

(b) Explain assignable an unassignable causes of variability in quality

(c) Explain working of Auto collimators. 2x8=16

$f\delta\theta = \Delta S$

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Roll No. 22UMEC1134

BE III (ME)

3

IE

THIRD B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(VI SEMESTER)

ME 353 A: INDUSTRIAL ENGINEERING (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) An Asset is purchased for Rs. 5,50,000 with an estimated life of 15 years its scrap value is estimated as Rs. 50,000. What will be the depreciation charged in the 5th year and book value at the 5th year? Assume a rate of return of 10% per year. 8

wrong

(b) Explain the different steps in value Engineering Process? Explain the Methodology? 8

2. (a) Explain the theory of Demand and Supply? Make a relationship chart between Demand and Supply? 8

(b) What is forecast Error? Explain the various types of Error used in forecasting? 8

3. (a) What do you understand by term budget? What necessary action should be taken for control of Budget? Explain by example. 8

(b) What are the commonly used qualitative forecasting techniques? Explain them. 8

4. (a) Explain the different types of cost involved in Production? Briefly explain them by suitable example. 8

(b) Explain the product life cycle? With suitable diagram. 8

5. (a) What is Ergonomics? Why it is required? 8

(b) What is maintenance? Classify it. Explain each type with suitable example. 8

6. (a) What are the methods of wage payment? Is wage payment may be a reason of Industrial disputes? 8

(b) Explain them :

(i) Industrial Relations

(ii) Trade Unions

(iii) Waste Management

(iv) Legal legislation and acts 8

7. Write short notes (Any four)

(a) Reliability

(b) Time Study

(c) Corrective Maintenance

(d) Break - down

(e) Work Sampling

(f) Exponential Smoothing Method. 4X4=1

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BE III (ME) **Roll No.** 22UMEC1134

3

Mec. Vib. (M)

THIRD B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(VI SEMESTER CBCS)

ME 354A : MECHANICAL VIBRATIONS (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) All questions carry equal marks.

(iii) Answer should be to the point; assume suitable data, if not given and mention it clearly.

(iv) Draw neat sketches, wherever necessary.

1. (a) Add the following harmonic motions analytically and check the solution graphically to determine magnitude and phase difference of it. $x_1=4\cos(\omega t+10)$ and $x_2=6\sin(\omega t+60)$. 8

(b) Find the equivalent spring constant of cantilever beam subjected to a concentrated load F 100 kN at its end.

Cantilever beam is made of mild steel having Young's modulus E = 200 GPa, density $\rho = 7800\ kg/m^3$, where the cross-section of beam is rectangular with width b=600 mm and thickness h = 4 mm, length of beam is L=1000mm. Also, determine the natural frequency of beam. 8

2. (a) A torsional pendulum when immersed in oil indicates its natural frequency as 320 Hz. But when it was put to vibration in vacuum having on damping, its natural frequency was observed as 375 Hz. Find the value of damping factor of the oil. 8

(b) Explain the force transmissibility for the damped single degree of freedom system with vector diagram; also explain transmissibility verses frequency ratio for different amount of damping. 8

3. (a) A gun barrel having mass 560 kg is designed with the following data. Initial recoil velocity 36 m/sec and Recoil distance on firing 1.5m. Calculate the following: (a) spring constant, (b) damping coefficient and (c) time required for the barrel to return to a position 0.12 m from its initial position. 8

(b) A vibrating system is defined by the following parameters: m = 3kg, k = 100 N/m, c=3N-sec/m. Determine (a) the damping factor, (b) the natural frequency of damped vibration, (c) logarithmic decrement, (d) the ratio of two consecutive amplitudes and (c) the number of cycles after which the original amplitude is reduced to 20 percent. 8

4. (a) Find the natural vibration frequencies and mode shapes for the torsional system as shown in Fig. 1. Here, diameter of rod 1 and rod 2 is 55 mm and 35mm, respectively; and length of rod 1 and rod 2 is 1.0 m and 1.2m, respectively, Both shafts are made of mild steel. Whereas masses of disc 1 and disc 2 is 1.5 kg and 0.8 kg, respectively and radius of rotation is 0.4 m and 0.2m, respectively.

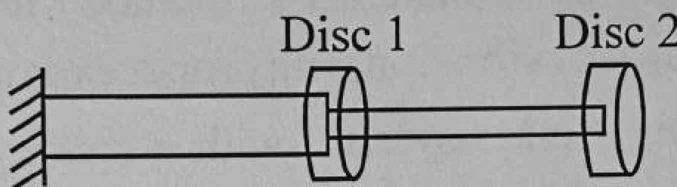

Fig. 1

8

(b) Determine the two natural frequencies and the corresponding mode shapes for the system as shown in Fig. 2. The string is stretched with a large tension T.

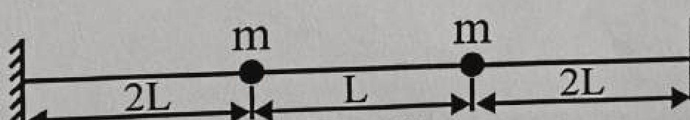

Fig. 2

8

5. (a) The system as shown in Fig. 3 is displaced from its static equilibrium position to the right a distance of 0.01 m. An impulsive force acts towards the left on the mass at the instant of its release to give it on initial velocity V_0 in that direction. If the system has the following parameters : k= 15700 N/m, c= 1570 N-sec/m and m=9.8 kg.

(i) Derive the expression for the displacement from the equilibrium position in terms of time t and initial velocity V_0.

(Contd.)

(ii) What value of V_0 would be required to make the mass pass the position of the static equilibrium / 100 second after it is applied?

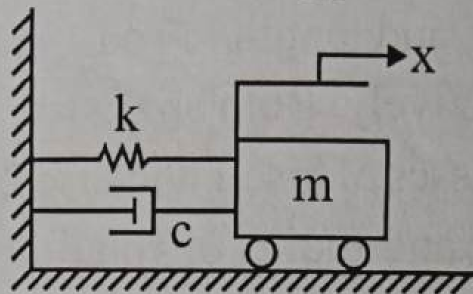

Fig. 3

8

(b) Explain the Dunkerley's method for finding the frequency of vibration with proper example. 8

6. (a) A spring - mass system, with a spring stiffness 9500 N/m, is subjected to a harmonic force of magnitude 100N and frequency 25Hz. The mass is found to vibrate with an amplitude of 0.2m. Assuming that vibration starts from rest ($x_0 = v_0 = 0$), determine the mass of the system.

8

(b) A block of mass 0.05 kg is suspended from a spring having a stiffness of 25 N/m. The block is displaced downwards from its equilibrium position through a distance of 2 cm and released with an upward velocity of 3 cm/sec. Determine (a) natural frequency, (b) period of oscillation (c) maximum velocity (d) maximum acceleration (e) phase angle. 8

7. Write short notes (any four). 4x4=16

(a) Explain the whirling of shaft and derive the formula for transverse deflection of shaft.

(b) Explain the principles of vibrometer and an accelerometer and write difference in both.

(c) Explain the Holzer method.

(d) Explain the principle of undamped dynamic vibration absorber.

(e) Explain the Equivalent circuits system with neat sketch.

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Roll No. 22UMEC1134

BE III (ME)

3

MD

THIRD B.E. EXAMINATION - 2024
MECHANICAL ENGINEERING
(VI SEMESTER CBCS)
ME 355 A: MACHINE DESIGN III (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

(iv) Use of design data hand book is permitted.

1. (a) Mention design considerations for a piston. 6

(b) Explain the buckling consideration in connecting rod. 10

2. (a) Explain significance of lubrication in roller bearing. 6

(b) Explain pressure distribution in Journal bearing. 10

3. Design a journal bearing for a centrifugal pump from the following data:

Load on the journal = 20 000 N; Speed of the journal = 900 r.p.m.; Type of oil is SAE 10, for which the absolute viscosity at 55°C = 0.017 kg/m-s; Ambient temperature of oil = 15.5°C, Maximum bearing pressure for the pump = 1.5 N/mm^2. Calculate also mass of the lubricating oil required for artificial cooling, if rise of temperature of oil be limited to 10°C. Heat dissipation coefficient = 1232 W/m^2/°C. 16

4. A four stroke diesel engine has the following specifications : Brake power = 5 kW; Speed = 1200 r.p.m. ; Indicated mean effective pressure = 0.35 N/mm^2; Mechanical efficiency = 80%

Determine : 1. bore and length of the cylinder; 2. thickness of the cylinder head; and 3. size of studs for the cylinder head. 16

5. The maximum fluctuation of energy for a multicylinder engine is 2475 N-m. The engine speed is 900 r.p.m. and the fluctuation in speed is not to exceed 2% of the mean speed. Find the mass and cross-section of the flywheel rim having 650 mm mean diameter. The density of the material of the flywheel may be taken as 7200 kg/m^3. The rim is rectangular with the width 2 times the thickness. Neglect effect of arms, etc. 16

6. Design a connecting rod for an I.C. engine for the following data:

Speed : 1600 r.p.m.

Maximum pressure of 2.8 N/mm^2.

Piston diameter: 80 mm

Mass of the reciprocating parts per cylinder: 1.95 kg

Length of connecting rod : 350 mm

Stroke of piston : 200 mm

Compression ratio 7 : 1.

Assume suitable factor of safety for the design.

Length to diameter ratio for big end bearing : 1.2

Length to diameter ratio for small end bearing : 2

Bearing pressures as 10 N/mm^2 and 15 N/mm^2 for big and small end bearings. 16

7. Select a single row deep groove ball bearing with the operating cycle listed below, which will have a life of 15 000 hours.

Fraction of cycle	Type of load	Radial (N)	Thrust (N)	Speed (rpm)	Service factor
1/10	Heavy shocks	2000	1200	400	3
1/10	Light shocks	1500	1000	500	1.5
1 / 5	Moderate shocks	1000	1500	600	2
3/5	No shock	1200	2000	800	1

Assume radial and axial load factors to be 1.0 and 1.5 respectively and inner race rotates. 16

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BE III (EE)

1

Open Elective-III

THIRD B.E. EXAMINATION, 2024

ELECTRICAL ENGINEERING

(VI SEMESTER)

EE 391 A : OPEN ELECTIVES-III

INTELLIGENT COMPUTING TECHNIQUES

Time – Three Hours

Maximum Marks – 100

Note: (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Define Artificial Intelligence. Discuss the important attributes of an AI System. 10

(b) Describe the organisation and functioning of Expert System building tools. 10

2. (a) Compare the different forms of knowledge representation techniques. 10

(b) Draw the semantic network for representing the knowledge about Open Elective EE 391A in the Academic Structure of MBM University. 10

3. (a) Explain the concept and application of the Backward and Forward Chaining control strategies. 10

Contd.

(b) Explain the hill climbing search method and its limitations. 10

4. (a) Explain the process of Non Monotonic Reasoning. 10

(b) Describe the various forms of analogical learning along with example of each of them. 10

(a) Explain the functioning of Multi-layer Perceptions and their learning algorithm. 10

(b) What is Self-Organisation? Explain its importance and application with a suitable example. 10

(a) Explain the concepts of fuzzification and defuzzification and various methods used for implementing them. 10

(b) Discuss the different forms of soft computing along with their applications. 10

Write short notes on any TWO of the following:- (10x2=20)

(a) Statistical Reasoning

(b) Conceptual Dependencies

(c) Learning in Neural Networks.

MBM University, Jodhpur

Department of Mechanical Engineering

BE VIIth Sem (Ist Class Work Sessional) Exam 2024-25

Subject: Heat & Mass Transfer – I Paper Code: ME 401A
Time: 1 Hr MM: 20 Marks

Q.1/CO1: What are the different modes of heat transfer? Explain their differences with the help of governing laws. (6)

Q.2/CO2: What do you mean by critical radius of insulation? (4)

Q.3/CO2: A 0.8 m high, 1.5 m wide double pane window consists of two 4 mm thick layers of glass (k = 78 W/mK) and is separated by a 10 mm wide stagnant air space (k = 0.026 W/mK). The room is at 20°C and the outside air is at -10°C. The heat transfer coefficients are h_i = 10 and h_o = 40 W/m^{2}K. Find the rate of heat transfer through the window and the inside surface temperature. (4)

Q.4/CO2: A steam pipe made of steel (k = 58 W/mK) has I.D. of 160 mm and O.D. of 170 mm. The saturated steam flowing through it is at 300°C, while the ambient air is at 50°C. It has two layers of insulation, the inner layer (k = 0.17 W/mK) is 30 mm thick and the outer layer (k = 0.02[illegible] W/mK) is 50 mm thick. The heat transfer coefficients on the inside and outside walls are 30 and 5.8 W/m2K respectively. Find the rate of heat loss per unit length of the pipe. (6)

MBM University, Jodhpur

Department of Mechanical Engineering

B.E. VIIth Semester (2nd Class Work Sessional) Exam 2024-25

Subject: Heat & Mass Transfer – I Paper Code: ME 401A
Time: 1 Hr MM: 20 Marks

Q.1/CO3: Derive expressions for temperature distribution and heat dissipation in a straight fin insulated at the tip of rectangular profile. (5)

Q.2/CO4: A solid copper sphere of 10 cm diameter [ρ = 8954 kg/m3, C_p = 383 J/kg K, k = 386 W/m K], initially at a uniform temperature of 250°C, is suddenly immersed in a well-stirred fluid which is maintained at a uniform temperature 50°C. The heat transfer coefficient between the sphere and the fluid is h = 200 W/m^2 K. Determine the temperature of the copper block at 5 min after the immersion. T = 120 or 121 (5)

Q.3/CO5: Show that the emissive power of a black body is π-times the intensity of emitted radiation. (4)

Q.4/CO6: A small sphere (outside diameter = 60 mm) with a surface temperature of 300°C is located at the geometric centre of a large sphere (inside diameter = 360 mm) with an inner surface temperature of 15°C. Calculate how much of emission from the inner surface of the large sphere is incident upon the outer surface of the small sphere; assume that both sides approach black body behaviour. What is the net interchange of heat between two spheres? (6)

Session : 2024-25
Date :- 14/11/24
Time :- 10:30 AM

Second Midterm Exam (B.E. Mech, VII Sem)

Subject: Manufacturing Technology ME 404 A

Note: Attempt questions summing up to total 20 marks

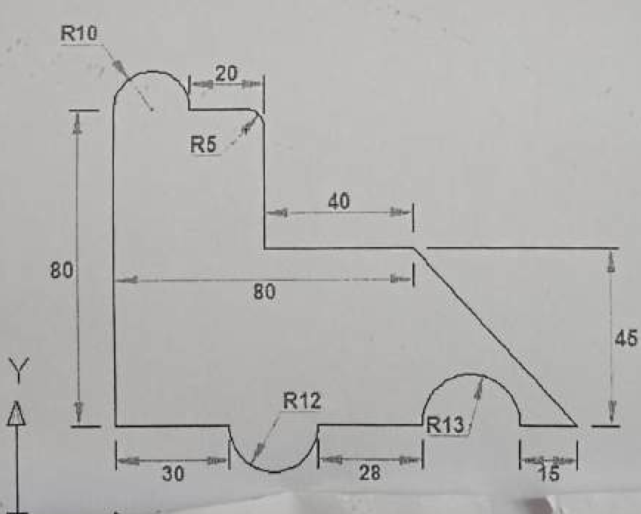
1. An electric discharge machining operation is performed on tungsten and zinc. Determine the amount of both metal removed in the operation after 1 hour at a discharge current of 20 amps. Use metric units and express the answers in mm3/hr. (Melting temperature of tungsten is 3410°; Melting temperature of zinc is 420°) **5 marks**

2. The frontal working area of the electrode is 3.7 in^2 in an ECM operation on pure aluminum. Applied current = 1500 amps, and voltage = 12 volts. The specific removal rate of aluminum is 1.26×10^{-4} in^3/amp-min. (a) If the ECM process is 85% efficient, determine the rate of metal removal in in^3/hr. (b) If the resistivity of the electrolyte = 6.8 ohm-in, determine the working gap. **5 marks**

3. Compare Electrochemical Machining and Electric Discharge Machining in terms of operating principle, key elements and process parameters. **10 marks**

4. Write about the architecture of NC systems and classify the control loops used in NC machines. Differentiate between manual and computer assisted part programming. **10 marks**

5. Write CNC part program for the following objects. Assume appropriate feed rate and cutting speed. **10 marks**



First Midterm Exam (B.E. Mech, VII Sem) 2024-25

Subject: Manufacturing Technology ME 404 A

Time: 1 Hour **Max Marks: 10**

1. Indicate the mathematical equation for the flow curve. How does increasing temperature affect the parameters in the flow curve equation? Compare hot working and cold working in terms of process, products produced and resulting properties. 2
2. A single-pass rolling operation reduces a 20 mm thick plate to 18 mm. The starting plate is 200 mm wide. Roll radius = 250 mm and rotational speed = 12 rev/min. The work material has a strength coefficient = 600 MPa and a strength coefficient = 0.22. Determine (a) roll force, (b) roll torque, and (c) power required for this operation. 4
3. In blanking of a circular sheet-metal part, is the clearance applied to the punch diameter or the die diameter? A blanking operation is to be performed on 2.0-mm thick cold-rolled steel. The part is circular with diameter = 75.0 mm. Determine the appropriate punch and die sizes for this operation. 4

BE IV Mid (7^{th}) Sem Exam 2024-25 Max Marks: 20

ME-403A: Hydraulic Machines (M) Time - 1:00 hr

3. a. What are the hydraulic machines, classify them and derive the relationship of specific speed for a turbine and a pump. 4

   b. Calculate the power given by a jet of water to runner of Pelton wheel having tangential velocity 21 m/s, net head 52 m and discharge rate of 0.05 m^3/s. Take side clearance angle of 15° and C_v as 0.98. 6

4. a. For a series of curved vanes mounted radially on the periphery of a wheel, show that degree of reaction is given by 6

$$R = 1 - [(V_1^2 - V_2^2)/2g.H_e] \quad \text{with usual notations}$$

   c. A lawn sprinkler with two nozzles of 5 mm diameters each at 20 cm and 15 cm radii is connected to a tap discharges water at 6 L/min. What torque will the sprinkler exert on hand if (i) held stationary and (ii) at angular velocity will it rotate freely. 4

BE IV Mid (7^{th}) Sem Exam 2024-25 Max Marks: 20

ME-403A: Hydraulic Machines (M) Time - 1:00 hr

1. Prove that in a double acting reciprocating pump, the work saved by fitting air vessels is about 39.2%. 5
2. A centrifugal pump is designed to run at 950 rpm to deliver 0.4 m^3/s to a head of 16 m. If the pump is coupled to a motor of speed 1450 rpm. Calculate the discharge, head and power input. Assume, overall efficiency is 0.82 remains constant. 5
3. What is the difference between a fluid coupling and fluid torque converter? Obtain an expression for the efficiencies for the hydraulic ram. 5
4. The inlet of a draft tube for reaction turbine is 2.5 m above tail race level. The outlet area is 3 times to inlet area, velocity at inlet is 8 m/s and kinetic head recovery is 80%. Considering atmospheric head as 10 m of water column, determine pressure at draft tube inlet. 5

DEPARTMENT OF MECHANICAL ENGINEERING
MBM UNIVERSITY, JODHPUR
MID TERM-1:-ME 405 A: Operation Research

ME: 1 Hr.

MM: 20

1. In a public telephone booth, the arrivals on an average are 15 per hour. A call on an average takes three minutes. If there is just one phone, find i) the expected number of callers in the booth at any time ii) the proportion of the time, the booth is expected to be idle. 5

2. The demand for an item is 18000 units per year. The holding cost is Rs1.20 per unit time and the cost of shortage is Rs.5.00. The production cost is Rs.400.00. Assuming that replacement rate is instantaneous determine the optimum order quantity. 5

3. A small project consists of seven activities for which the relevant data are given below:

Activities	Preceding Activity	Duration(in days)
A	-	4
B	A	7
C	-	6
D	C	5
E	B	7
F	D,E	6
G	F	5

Draw the network and find the project completion time. 10

DEPARTMENT OF MECHANICAL ENGINEERING
MBM UNIVERSITY, JODHPUR

MID TERM-1:-ME 405 A: Operation Research

TIME : 1 Hr.

MM: 20

Note: All questions are of equal marks.

1. Let us solve the simplex method following problem

Maximize: $Z = 3X_1 + 2X_2$

Subject to: $-X_1 + 2X_2 < 4$
$3X_1 + 2X_2 < 14$
$X_1 - X_2 < 3$
$X_1, X_2 > 0$

10

2. The handy- dandy company wishes to schedule the production of a kitchen appliance that requires two resources- labor and material. The company is considering three different models and its production engineering department has furnished the following data:

	Model		
	A	B	C
Labor(hrs per unit)	7	3	6
Materials (pounds per unit)	4	4	5
Profit ($ per unit)	4	2	3

The supply of raw material is restricted to 200 pounds per day. The daily availability of labor is 150 hours. Formulate a linear programming model to determine the daily production rate of the various models in the order to maximize the total profit. 10

Attempt any two Question

(1) Explain Combustion in S.I engine. Draw P-θ diagram and explain three stages of combustion

(2) Discuss the phenomenon of knocking and the various engine factors which affect knocking in SI engine

(3) What are fuel-air mixture requirement for following operation in SI engine

(a) Idling and low load

(b) Cruising range

(c) Maximum Power range

---

(4) A simple Carburettor is required to supply 4.5 kg air and 0.5 kg of fuel per min. The fuel specific gravity is 0.8. The air is initially at 1 bar, 300 K. Calculate throat diameter of choke for flow velocity 110 m/s If the pressure drop across the fuel metering orifice is 0.85 of that choke. Calculate orifice diameter

C_d (venturi) = 0.8

C_d (fuel orifice) = 0.6

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Roll No. 22UMEC1139

BE IV (ME)

3

HMT - I

FINAL B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(VII SEMESTER CBCS)

ME - 401 A : HEAT & MASS TRANSFER - I

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Explain the resistance concept to illustrate the analogy of heat flow and the flow of electricity. 6

   (b) What is the significance of Thermal Diffusivity? 2

   (c) Show that the temperature profile for heat conduction through a cylindrical wall having uniform thermal conductivity is logarithmic. 8

2. (a) Derive general heat conduction equation for isotropic material in Cartesian co-ordinates. 8

(b) A wall is constructed of several layers. The first layer consists of masonry brick (K= 0.66 w/mk) of $R_T = 1.367$
25 cm thick, the second layer of 2.5 cm thick $U_1 = 0.73$
mortar (K= 0.7 w/mk), the third layer of 10 cm thick limestone (K= 0.66 w/mk) and the outer layer consists of 1.25 cm thick plaster (K= 0.7 w/mk). The heat transfer coefficient on interior and exterior of the wall fluid layer are 5.9 w/m^2k and 1.16 w/m^2k respectively. Find (i) over all heat transfer coefficient from air on the interior to air at exterior on the wall.

(ii) Rate of heat transfer per unit area if the interior of room is 26°C while outside air temperature is -7°C. 24.14

(iii) Temperature at the junction between the mortar and limestone. 281.2 8

3. (a) Derive the expressions for temperature distribution and heat transfer for hollow cylinder with uniform heat generation, assuming one dimensional conduction heat transfer. 8

(b) In a metallurgical industry for mass annealing process, two thousand spheres made of copper of diameter 4 mm are annealed in a furnace. The $b = 0.017$
initial temperature of the spheres is 25 °C and the $t = 64.89$ sec
temperature of furnace is 500 °C. The heat transfer

coefficient is 35 w/m^2k, specific heat is 335 J/kgk and density is 8800 kg/m^3. Calculate the time required for the sphere to reach the temperature of 350 °C. 8

4. (a) Discuss the criteria of selection of fins. What is the difference between the Fin effectiveness and the Fin efficiency? 6

(b) The impeller of a centrifugal pump is mounted on a horizontal shaft of 35 mm diameter and 600 mm length. It has its first bearing located at 110 mm. The temperature of impeller is 550 °C. The ambient air temperature is 30 °C and the heat transfer coefficient is 19 w/m^2k. The thermal conductivity of steel shaft is 20 w/mk. Compare the value of temperature at the bearing under the following conditions:-

(i) Infinite length of the shaft

(ii) Insulated end and

(iii) Convective heat transfer from the end. 10

5. (a) Explain Kirchhoff's law. How does an enclosure with a small hole in it behave as a black body? 8

(b) In process of estimating the emission from the sun, it may be treated as a black body with a surface temperature of 5800 K at a mean distance of 15×10^{10} m from the earth. The diameter of the Sun is 1.4×10^9 m and that of Earth is 12.8×10^6 m. Estimate the following:-

(i) Total energy emitted by the sun.

(ii) Emission received per m^2 just outside the earth's atmosphere.

(iii) Total energy received by the earth if no radiation is blocked by earth's atmosphere. 8

6. (a) Explain the "surface resistance" and "space resistance". How can you construct a radiation network for two gray surfaces exchanging radiant energy? 8

(b) Consider two large parallel plates one at t_1,= 727°C with emissivity $\varepsilon_1 = 0.8$ and other at $t_2 = 227$ °C with emissivity $\varepsilon_2 = 0.4$. An aluminium radiation shield with an emissivity, $\varepsilon = 0.05$ on both side is placed between the plates. Calculate the percentage reduction in heat transfer rate between the two plates as a result of the shield. 8

Gate 1995 ME

0.065

93.41

7. Write the short notes on the following:-

(i) Critical radius of insulation. → k/h

(ii) Lumped heat capacity analysis.

(iii) Planck's Law.

(iv) Shape Factor. 4x4=16

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Roll No. 22UMEC1134

BE IV (ME)

3 MT

FINAL B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(VII SEMESTER CBCS)

ME - 404 A : MANUFACTURING TECHNOLOGY

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) The data obtained from a rectangular strain gauge rosette attached to a stressed steel member are ε_o = -220×10^{-6}, $\varepsilon_{45} = 120\times10^{6}$, and $\varepsilon_{90} = 220\times10^{-6}$. Given that the value of $E = 2\times10^{5}$ N/mm^2 and Poisson's Ratio $\mu = 0.3$, calculate the values of principal stresses acting at the point and their directions. 8

(b) What is the difference between engineering stress and true stress in a tensile test? Explain work hardening. 8

2. (a) A single-pass rolling operation reduces a 20 mm thick plate to 18 mm. The starting plate is 200 mm wide. Roll radius = 250 mm and rotational speed = 12 rev/min. The work material has a strength coefficient = 600 MPa and a strain hardening exponent = 0.22. Determine

(i) Roll force

(ii) Roll torque

(iii) Power required for this operation. 8

(b) What is extrusion? Distinguish between direct and indirect extrusion. 8

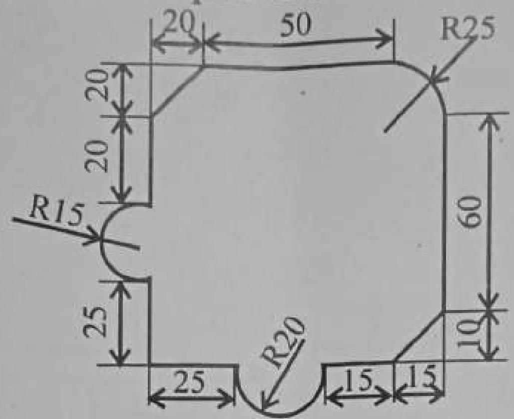
3. (a) What is the difference between blanking and punching operation? Explain different defects in drawn sheet metal parts. 8

(b) Explain with a neat sketch the principle and working of Electro-Chemical Machining (ECM) process. Give also its advantages, disadvantages and applications. 8

4. (a) Explain the working principle of Abrasive Jet Machining method. State its advantages and disadvantages. 8

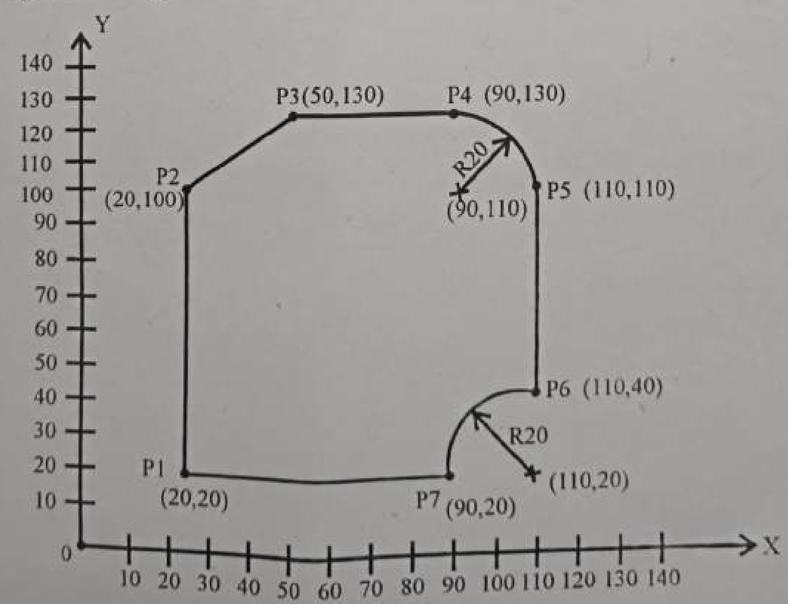
(b) Explain with neat sketches the 'thread rolling' method of making threads. State its advantages and limitations. 8

5. (a) Write a CNC Milling program for the given component using subroutines. Profile milling depth = 6 mm. Take the depth of cut = 0.5 mm, speed = 1200 rpm. Assume suitable feed.



(b) Give the comparison of 'Gear hobbing' and 'Gear shaping'. Discuss briefly the sources of errors in manufacturing gears. 8

6. What do you mean by APT? Write the APT program for the following milling operation on End Mill. Given that cutter diameter = 20 mm, spindle speed = 50 RPM and Feed rate = 2 mm/rev.



16

7. Write short note on any of two of the following :-

(a) Friction and lubrication in metal forming.

(b) High energy rate forming.

(c) Types of Robots.

2x8=16

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Roll No. 22UMEC1B.4

BE IV (ME)

3

HM

FINAL B.E. EXAMINATION - 2025
MECHANICAL ENGINEERING
(VII SEMESTER CBCS)
ME-403 A: HYDRAULIC MACHINES

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Define moment of momentum equation. Also derive the expression for determining the force exerted by a flowing fluid on a pipe bend. 6

done

PN-289

(b) A 45° reducing bend is connected in a pipe line, the diameters at the inlet and outlet of the bend being 60 cm and 30 cm respectively. Find the force exerted by water on the bend if the intensity of pressure at inlet to bend is 8.829 N/cm^2 and rate of flow of water is 600 Litres/sec. 10

Que 6.29

Same

P = 5.38×10^4
Fx = 13586.56
Fy = − 6799.5 F = 14,976

2\. (a) Show that the hydraulic efficiency for a Francis turbine having velocity of flow through runner as constant, is given by the relation.

$$\eta_h = \frac{1}{1 + \frac{\frac{1}{2}\tan^2\alpha}{\left(1 - \frac{\tan\alpha}{\tan\theta}\right)}}$$

Where $\alpha =$ Guide blade angle

$\theta =$ Runner vane angle at inlet.

The turbine is having radial discharge at outlet. 8

(b) A Pelton wheel develops 8000 KW under a net head of 130 m at a speed of 200 rpm. Assuming the coefficient of velocity for the nozzle 0.98, hydraulic efficiency 87%, speed ratio 0.46 and jet diameter to wheel diameter ratio 1/9, determine:

(i) The discharge required. 84.40 m³/s
2.173 m

(ii) The diameter of the wheel. d = 0.241

(iii) The diameter and number of jets required.
24

(iv) The specific speed 0.1924 (31)

Take mechanical efficiency is 0.75. 8

3\. (a) What is a draft-tube? Why is it used in reaction turbine? Describe with sketch two different types of draft-tubes. 8

(b) A Kaplan turbine runner is to be designed to develop 7.357 MW shaft power. The net available head is 550 cm. Assume that the speed ratio is 2.09 and flow ratio is 0.68, and the overall efficiency is 60%. The diameter of the boss is $1/3^{rd}$ of the diameter of the runner. Find the diameter of the runner, its speed and its specific speed. 8

4. (a) Define Cavitation. What are the effects of Cavitation? Give the necessary precautions against Cavitation.

Occe 19.15 Exme

(b) A Centrifugal pump with 120 cm diameter runs at 200 rpm and pumps 1880 Litres/sec, the average lift being 600 cm. The angle which the vanes make at exit with the tangent to the impeller is 26° and the radial velocity of flow is 2.5 m/sec. Determine the manometric efficiency and the least speed to start pumping against a head of 600 cm, the inner diameter of the impeller being 60 cm. 8

Occe 20.10

5. (a) In a single - acting reciprocating pump, stroke length of 15 cm. The suction pipe is 700 cm long and the ratio of the suction diameter to the plunger diameter is 0.75. The water level in the sump is

(Contd.)

250 cm below the axis of the pump cylinder, and the pipe connecting the sump and pump cylinder is 7.5 cm diameter. If the crank is running at 75 rpm, determine the pressure head on the piston:

(i) In the beginning of the Suction Stroke.

(ii) In the end of the Suction Stroke.

(iii) In the middle of the Suction Stroke.

Take Co-efficient of friction as 0.01 12

(b) What is an Air Vessel? Describe the function of the air vessel for reciprocating pumps. 4

6. An inward flow reaction turbine has external and internal diameters are 0.9 m and 0.45 m respectively. The turbine is running at 200 r.p.m. and width of turbine at inlet is 200 mm. The velocity of flow through the runner is constant and is equal to 1.8 m/sec. The Guide blades make an angle of 10° to the tangent of the wheel and the discharge at the outlet of the turbine is radial. Draw the inlet and outlet velocity triangles and determine:

(i) The absolute velocity of water at inlet of runner.

(ii) The velocity of whirl at inlet.

(iii) The relative velocity at inlet.

(iv) The runner blade angles

(v) Width of the runner at outlet.

(vi) Mass of water flowing through the runner per second.

(vii) Head at the inlet of the turbine.

(viii) Power developed and hydraulic efficiency of the turbine. 16

7. Write short notes on the following with suitable sketches as needed : (Any four)

(i) Hydraulic ram

(ii) Hydraulic torque converter

(iii) Hydraulic Intensifier

(iv) Pump Characteristics

(v) Gear pump.

(vi) Hydraulic coupling

4x4=16

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Roll No. 22UMEC1184

BE IV (ME)

3 **OR**

FIRST B.E. EXAMINATION - 2025
MECHANICAL ENGINEERING
(VII SEMESTER CBCS)
ME - 405 A: OPERATIONS RESEARCH

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1\. (a) Let us solve by the Simplex Method the following problem :-

Minimize : $Z = -3x_1 + x_2 + x_3$

Subject to : $x_1 - 2x_2 + x_3 \leq 11$

$-4x_1 + x_2 + 2x_3 \geq 3$

$2x_1 - x_3 = -1$

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$ 12

(b) Write the special cases of LPP

(i) Degeneracy

(ii) Infinite Solution

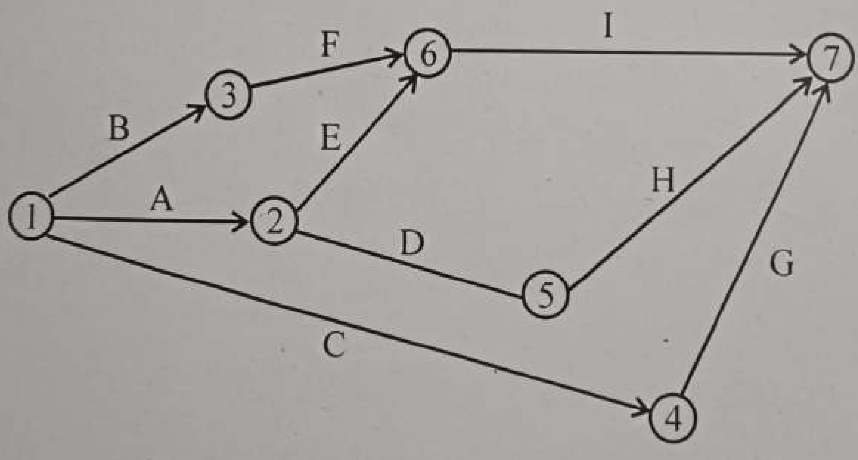
(iii) Infeasibility

(iv) Unbounded Solution 4

2. (a) Write the necessary steps for making the Network diagram? 4

(b) An Architect's project have following activities in given Network diagram

Activities	A	B	C	D	E	F	G	H	I
Optimistic time t_o	5	18	26	16	15	6	7	7	3
Pessimistic time t_p	10	22	40	20	25	12	12	9	5
Most likely time t_m	8	20	33	18	20	9	10	8	4



Determine the following

(a) Expected completion time 47.8 weeks

(b) Critical path – 1-4-7

(c) Slack for each activity

(d) The probability of expected completion time of the project if the original schedule time of completing project is 41.5 weeks. 12

-0.015

Jodhpur Rajasthan Dec 19, 2024, 07:32

3. (a) What do you understand by Economic Order Quantity? Explain the Wilson - Harris Model? 4

(b) Annual Demand is 8000 unit ordering cost C_o is 1800 per/order holding cost is 10% of unit purchase price of Inventory. Determine the best order size when the items can be purchase as given below.

Lot size	Unit price
1-999	220
1000-1499	200
1500-1999	190
2000 & above	185

12

4. (a) A train reservation facility as five counter each capable of handling 20 request/hr the person's arrive at a mean rate of 90/hr. Calculate

(i) Average Number of person's at any time at this facility.

(ii) The mean time spend by a person at the facility

(iii) Average length of queue at each counter

(iv) What would happen to queue if the arrival rate reaches 105/hr 8

(b) Solve the following Assignment Problem.

Machines

Jobs		M_1	M_2	M_3	M_4	M_5
	J_1	10	11	4	2	8
	J_2	7	11	10	14	12
	J_3	5	6	9	12	14
	J_4	13	15	11	10	7

Assign the appropriate job on Machine. 28 8

(Contd.)

5. (a) Solve the Travelling Salesmen problem given by

	A	B	C	D	E
A	∞	2	5	7	1
B	6	∞	3	8	2
C	8	7	∞	4	7
D	12	4	6	∞	5
E	1	3	2	8	∞

15

8

(b) Solve the following Transportation Problem.

	D_1	D_2	D_3	D_4	Availability
O_1	1	2	1	4	30
O_2	3	3	2	1	50
O_3	4	2	5	9	20
Requirement	20	40	30	10	

8

6. (a) Solve the following game using the Dominance concept

Player B

		B_1	B_2	B_3
Player A	A_1	4	5	8
	A_2	6	4	6
	A_3	4	2	4

$\frac{14}{3} \to V$

$\frac{2}{3}, \frac{1}{3} \to A$

$\frac{1}{3}, \frac{2}{3}$ B

8

(b) Solve the following Linear Programming Problem for Maximization of Profit by graphical method.

$$\text{Max } z = 30x_1 + 20x_2$$

sub to .

$$2x_1 + x_2 \le 90$$

$$x_1 + 2x_2 \le 80$$

$$x_1 + x_2 \le 50$$

$$x_1, x_2 \ge 0$$

1400

8

7. Write short notes (any four)

(a) Duality

(b) Simulation

(c) Sensitivity Analysis

(d) Difference in PERT & CPM

(e) Customer's attitute $4x4=16$

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Roll No. 22UMEC1134

BE IV (ME)

3 **IC Engine**

FINAL B.E. EXAMINATION - 2025
MECHANICAL ENGINEERING
(VII SEMESTER CBCS)
ME - 402 A: INTERNAL COMBUSTION ENGINES

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Explain with p - V diagram the ideal and actual Otto-cycles using terms: time loss ratio, heat loss ratio, blow down loss ratio, work loss ratio, also compare them. 8

(b) A spark ignition engine is supplied with mixture of gasoline vapour C_8H_{18} and air. The exhaust analysis gives 13.5% CO_2. Calculate the fuel vapour to air ratio by volume and state whether it is a rich or lean mixture at complete combustion. 8

2. (a) What is abnormal combustion and its detrimental effects on engines performance? Explain with p - θ diagram the knock phenomena in S.I. engines and factors affecting the knock tendency. 10

(b) In a diesel engine air-fuel ratio is 30:1, air intake at 27°C and compression ratio is 16:1; calculate the ideal cycle efficiency? Calorific value of fuel is 42, 000 kJ/kg. 6

3. (a) What are the homogenous and heterogeneous mixtures? Explain the stages of combustion in CI engines with the help of P - θ diagram. 6

(b) The following data obtained during a test on four cyclinder, four stroke engine, bore = 100 mm; stroke length = 120 mm; speed = 1600 rpm; amount of fuel used = 0.2 kg/min; CV of fuel = 44000, kJ/kg; difference in tension on either side of brake pulley = 40 kg; brake cirumference = 300 cm. If the mechanical efficiency is 80%, calculate (i) brake thermal efficiency (ii) indicated thermal efficiency (iii) indicated mean effective pressure and (iv) break specific fuel consumption. 10

4. (a) Explain the process of scavenging with p - θ diagram in two storke engines and effect of operating parameters on volumetric efficiency with respective graphs. 6

(b) A diesel engine with cylinder bore of 10 cm has inlet valve of 4.3 cm and lift of 2.4 cm. The steady flow coefficient is 1.45 times the lift to valve diameter. If engine stroke length is 12.5 cm, determine the value of inlet valve Mach index at 2000 rpm. Is the inlet valve diameter adequate, if not, how large it should be? 10

5. (a) Why lubrication is required in I.C. engines, name different systems? Explain with neat sketch the construction and working of a dry-sump lubrication system. 6

(b) A two-stroke diesel engine of compression ratio 17 is running at piston speed of 12.7 m/s and scavenging pressure not to exceed 0.34 bar above atmosphere. The inlet air temperature is 34°C above atmosphere. If scavenging ratio is 1.0, determine the: (i) flow coefficient and (ii) indicated mean effective pressure with 75% compressor efficiency. Take atmospheric pressure of 1.0 bar at 15°C. 10

6. (a) What are the typical emissions and control systems for petrol and diesel engines; state their harmful effects on human, vegetation and environment? Explain with neat sketch the construction and working of electrostatic precipitator. 10

(b) What percentage of exhaust gas will be water vapour after combustion of octane fuel with 50% excess air at 260°C when cooled to 30°C at 1.0 bar? 6

7. Write short notes on the followings: 4X4=16

(i) Air pollution legislation in India and EURO standards

(ii) Octane and Cetane numbers for petrol and diesel fuels

(iii) Working of free piston engines and power output

(iv) Scavenging efficiency, trapping efficiency and perfect scavenging

MBM University, Jodhpur

Department of Mechanical Engineering

BE VIIIth Sem (1st Midterm Exam -2025)

Subject: Heat & Mass Transfer – II — Paper Code: ME 451A

Time: 1 Hr — MM: 20 Marks

Q.1 Show by dimensional analysis for force convection, $Nu = \phi$ (Re, Pr). (5)

Q.2 What is Stanton number? Explain Reynolds-Colburn analogy. (5)

Q.3 Air at 27 °C and 1 atm flows over a flat plate at a speed of 2 m/s. Calculate the boundary-layer thickness at distances of 20 cm and 40 cm from the leading edge of the plate. Assume that the plate is heated over its entire length to a temperature of 60∘C. Calculate the heat transferred in (a) the first 20 cm of the plate and (b) the first 40 cm of the plate. Properties of air at 43.5 °C are as $\upsilon = 17.36\times10^{-6}$ m^2/s, k = 0.02749 W/m· °C, Pr = 0.7, C_p = 1.006 kJ/kg.°C (5)

(a) $Re_x = 23041.5$
$\delta = 0.010m$
$h_x = 5.7$
$\bar{h}_x = 11.436$
$Q = 378.18$ W/m

(b) $Re = 0.46\times10^7$
$\bar{h} = 18.5$
$\frac{Q}{A} = 610.5$
$\delta = 0.0273m$

Q.4 Air is flowing over a flat plate 5 m long and 2.5 m wide with a velocity of 4 m/s at 15° C. If p = 1.208 kg/m^3 and v = 1.47 x 10-5 m^2/s, calculate: (i) Length of plate over which the boundary layer is laminar, and thickness of the boundary layer (laminar). (ii) Shear stress at the location where boundary layer ceases to be laminar, and (iii) Total drag force on the both sides on that portion of plate where boundary layer is laminar. (5)

$x = 1.83m$
$Re_x > 4.99\times10^5$
$\delta = 3.68$ m
$C_{FD} > 0.292$
$\bar{F}_x = 0.585$
$\tau_w = 5.74$
$F = 142.8$ N

First Midterm Exam (B.E. Mech, VIII Sem)

Subject: Production and Operations Management ME 454 A

Date: 05.02.2025 — **Maximum marks: 10**

1. Explain the various types of production systems in terms of characteristics, design and operational control with suitable examples. **3 marks**
2. Explain the various types of plant layouts along with their characteristics using suitable examples. **3 marks**
3. Daily production capacity of a bearing manufacturing company KC Mech corporation is 30000 bearings. The daily demand for the bearing is 15000. The holding cost per year of keeping a bearing in the inventory is Rs. 20. The setup cost for the production of a batch is Rs. 1800. What would be the economic batch quantity in number of bearing assuming 300 working days in a year? **2 marks**
4. A manufacturing company purchases 9000 parts of a machine for its annual requirements ordering for month usage at a time, each part costs Rs. 20. The ordering cost per order is Rs. 15 and carrying charges are 15% of the average inventory per year. You have been assigned to suggest a more economical purchase policy for the company. What advice do you offer and how much would it save the company per year? **2 marks**

MBM UNIVERSITY, JODHPUR 1124

1st Mid Term

Final Year VIII Semester 2024-25

ME452A: Gas Dynamics & Gas Turbines (M)

Date: 11/03/2024 | Exam Duration: 1 Hour | M.M.: 20

1. Draw the schematic diagram, P – V and T – S diagram of following: (09)
   a) Simple Cycle with Reheat and Heat Exchanger.
   b) Simple Cycle with Intercooler Heat Exchanger.
   c) Simple Cycle with Heat Exchanger, Reheater and Intercooler.
2. In a Gas turbine plant air enters the compressor at 1 bar and 280 K (1800). it is compressed to 4 bar with an iseutsopic efficiency of 80%. The maximum temperature at the inlet to the turbine is 83% (— 8°dC). The calorific value of the fuel used is 42 MJ/kg. calculate the following: (7 ½)
   a) Compressed work
   b) Heat supplied
   c) Turbine work
   d) Thermal efficiency
   e) Air/Fuel Ratio

   take $Cp_a = 1.005\ kJ/kg\ K$ $r_a = 1.4$ $C_{pg} = 1.145\ kJ/kg\ K$ $r_g = 1.33$
3. Determine the indicated mean effective pressure and efficiency of a Joule cycle if the temperature at the end of combustion is 1800°C and the temperature and pressure before compression is 340 K and 1 bar. The pressure ratio is 1.25. Assume $C_p = 1.005\ kJ/kg\ K$. (3 ½)

BE IV (8th Sem) Mid-Term Exam 2024-25 — Max Marks: 20

ME-453A: Power Generation (M) — Time - 1:00 hr

1. Why is electricity the most convenient form of energy? What is doubling time in electricity production? Review the development of power plants in India with their power shares and future possibilities. 5
2. State two basic parameters to be decided while designing a power station. Show that overall efficiency of a plant is the product of five component efficiencies. Explain the concept of power grids. 5
3. What is the common unit size of a steam power plant? State the selection criterions, working principle and advantages of supercritical boilers used in modern power plants. List the advantages of employing a condenser in thermal power plants. 5
4. What are the modern (i) fuel burners, (ii) dust collection methods, (iii) merits of burning the pulverised coal with limestone or dolomite (iv) ash handling systems? Explain with neat sketch the working of an electronic precipitator. 5

DEPARTMENT OF MECHANICAL ENGINEERING

MBM UNIVERSITY, JODHPUR

MID TERM-1:- ME 455 A: (b) FACILITIES LOCATION AND LAYOUT PLANNING

TIME: 1 Hr. | MM: 20

Note: All questions are of equal marks.

1. Write the criteria for site selection with suitable example ? 5
2. Write the factors that affect the site selection ? 5
3. Write the name of principles used in plant layout ? 5
4. Define the types of plant layout ? with example ? 5

MBM University, Jodhpur

Department of Mechanical Engineering

BE VIIIth Sem (2nd Midterm Exam -2025)

Subject: Heat & Mass Transfer – II Paper Code: ME 405A

Time: 1 Hr MM: 20 Marks

Q.1 What is the physical significance of Grashof number with reference to free convection heat transfer? What is Rayleigh number? (4)

Q.2 Explain the regimes of pool boiling. (5)

Ex 9.1 by Holman book

Q.3 A heat exchanger is to be designed to condense an organic vapour at a rate of 500 kg/min which is available at its saturation temperature 355 K. Cooling water at 286 K is available at a flow rate of 60 kg/s. The overall heat transfer hot fluid (vapour) coefficient is 475 W/m^2 ^{o}C. Latent heat of condensation of the organic vapour is 600 kJ/kg. Calculate: (a) The number of tubes required, if 25 mm outer diameter, 2mm thick and 4.87 m long tubes are available, and 487 (b) The number of tube passes, if the cooling water velocity (tube side) should not exceed 2 m/s. 1 (6)

Ex. 8.21 by R.k Rajput Book

Q.4 A vertical square plate, 30 by 30 cm, is exposed to steam at atmospheric pressure. The plate temperature is 98°C. Calculate the heat transfer and the mass of steam condensed per hour. Properties evaluated at the film temperature are: $\rho_l=960$ kg/m^3 $\mu_l=2.82\times10^{-4}$ kg/m·s, $k_l=0.68$ W/m·°C. At atmospheric pressure we have $T_{sat}=100$°C and $h_{fg}=2255$ kJ/kg. (5)

m = 3.70 kg/h

Q = 2367 W

DEPARTMENT OF MECHANICAL ENGINEERING

II MID TERM TEST VIII SEM

SUBJECT – GAS DYNAMICS AND GAS TURBINE

TIME - 1 Hour DATE -07/04/2025

Class Room

Q1. Explain different categories of compressible flow. (3)

Q2. Define Mach line ,Mach angle and Mach wave with sketch. (3)

Q3. For isentropic flow, derive the following relations: (8)

(a) $dA / A = (dP / \rho V^2)(1-M^2)$

(b) $T^*/T = \{2 / (\gamma+1)\} + \{(\gamma - 1) / (\gamma+1)\} M^2$

PYQ 2024

Q4. Air at a pressure of 0.4 MPa and temperature of 525 K flow with a velocity of 12000 m/min in a 325 mm diameter duct. Take Cp =1.005 KJ/kg-K, γ=1.4 Calculate: (6)

(a) Mas flow rate

(b) Stagnation temperature

(c) Mach number, and

(d) Stagnation pressure values assuming the flow as compressible and incompressible respectively.

BE IV (8^{th} Sem) Mid-Term-II Exam 2024-25

ME-453A: Power Generation (M)

Max Marks: 20

Time - 1:00 hr

1. Explain with sketch the working of a liquid metal-cooled FBR for power generation and fuel production. State problems in handling the nuclear fuel and waste management. 5

Ex 9.3 by P.K

2. Each fission of U^{235} yields 190 MeV energy. Assuming that 85% neutrons cause fission & rest produce isotope U^{236}; estimate fuel consumption/day to produce 3000 MW power. 5 3.9 ok

3. State the prime movers' selection criterions and what are the elements contribute to the electricity cost? Explain different tariff methods of charging the consumer. 5

Ex 1.9 by P.K

4. A 2000 MW thermal power plant have maximum and minimum demands of 1900 MW and 1200 MW in a year. Assuming load-duration curve to be a straight line over the period, calculate (i) load factor (ii) capacity factor and (iii) plant reserve factor. 5

$\frac{1550}{1900} = 0.81$ $\frac{1550}{2000} = 0.775$ 1.29 0.05

Second Mid-Term Exam 2024-25 (B.E. Mechanical, VIII Semester)

Subject: Production and Operations Management ME 454A **Date:** 08-04-25 **Total Marks:** 20

PYQ 2024 paper

Attempt any four questions

1. Explain Operations Scheduling with an example of processing 3 jobs considering following sequencing rules: (i) Shortest processing time (SPT), (ii) First come first serve (FCFS), (iii) Earliest due date (EDD). (5)
2. Explain Assembly Line Balancing with a suitable example. (5)
3. Give details of Parts Classification and Coding used in Group Technology. (5)
4. Explain Computer Aided Process Planning and differentiate between Retrieval and Generative CAPP (5)
5. Define Flexible Manufacturing System. Give details of components, functions and layouts of Flexible Manufacturing Systems. (5)

DEPARTMENT OF MECHANICAL ENGINEERING

MBM UNIVERSITY, JODHPUR

MID TERM-2

Subject:- ME 455 A: (b) FACILITIES LOCATION AND LAYOUT PLANNING

TIME: 1 Hr. **MM: 20**

Note: All questions are of equal marks.

Class

1. Draw flow chart for given example? Typist has to take dictation from the author's office. He has to type the letter, copy the letter and get letters attested by the author. He has to prepare the envelope and put these letters in the envelope. Finally, that letter to post office situated in front this office other side of road. Author's office is 20 feet away from typist office. **(10)**
2. Write the factors that affects the selection of material handling equipments? **(5)**
3. Write the classification of material handling equipments? **(5)**

NS

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BE IV (ME)

3

HMT - II

FINAL B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(MAIN/ATKT/EX)

(VIII SEMESTER)

ME 451 A: HEAT AND MASS TRANSFER - II (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Show by dimensional analysis for free convection

$$Nu = \phi(Gr, Pr)$$ 8

(b) Air at 20°C and a pressure of 1 bar is flowing over a flat plate at a velocity of 3 m/s. If the plate is 280 mm wide and at 56 °C, estimate the following quantities at x = 280 mm, given that the properties of air at the bulk mean temperature of 38 °C are: ρ=

δ = 0.018 m

Cf = 2.6×10^-3

Nu = 51.2

$1.1374\ kg/m^3$, $K = 0.02732$ w/mk $C_p = 1.005$ kJ/kgK, and $\vartheta = 16.768 X 10^{-6} m^2/s$:

(i) Boundary layer thickness (ii) Local friction coefficient (iii) Average friction coefficient (iv) Shear stress (v) Thermal boundary layer thickness (vi) Local convection heat transfer coefficient (vii) Rate of heat transfer by convection (viii) Total drag force on plate. 8

2. (a) Derive the two-dimensional momentum equation for the hydrodynamic boundary layer on a flat plate. 8

(b) In a power plant feedwater is flowing through a duct of rectangular cross-section 8 cm X 4 cm and the wall temperature is maintained at 170°C throughout. The feedwater flows at a rate of 300 kg/min, enters at a temperature of 20°C and is heated to 150 °C. Calculate the heat transfer coefficient and estimate the required length of the duet properties of water at 105 °C are $P_r = 1.64$, $\mu = 265 X 10^{-5}$ kg/ms, $C_p = 4.226$ kJ/kgK, $K = 683 X 10^{-6}$ kw/mk. 8

$\rho = ?$
$h = ?$
$Q = ?$

3\. (a) What do you mean by hydrodynamic and thermal entry lengths? 4

(b) What is the physical significance of Grashof number with reference to heat transfer by natural convection? What is Rayleigh number? 4

(c) A 0.15 m outer diameter steel pipe lies 2 m vertically and 8 m horizontally in a large room with an ambient temperature of 30 °C. The pipe surface is at 250 °C and has an emissivity of 0.60. Estimate the total rate of heat loss from the pipe to the atmosphere. The thermophysical properties at mean film temperature are: $\vartheta = 27.8 \times 10^{-6}$ m²/s, $P_r = 0.684$. 8

4\. (a) Explain the conditions under which dropwise condensation can take place. Why is the rate of heat transfer in dropwise condensation many times larger than in filmwise condensation? 4

(b) Explain why the condenser tubes are usually horizontal. 4

(c) Saturated steam at 110 °C condenses on the outside of a bank of 64 horizontal tubes of 25 mm outer diameter, 1 m long arranged in a 8x8 square array: Calculate the rate of condensation if the

$h = 6766 \times 20 = 4736$

tube surface is maintained at 100 °C. The properties of saturated water at 105 °C are $\rho = 954.7$ kg/m^3, K = 0.684 w/mk, $\mu = 271 \times 10^{-6}$ kg/ms and $h_{fg} = 2243.7$ kJ/kg. Had the condenser been vertical, what would have been the rate of condensation? $m = 1.6524 \times 18$ 8

5. X (a) Discuss the various regimes of boiling heat transfer. Comment on critical heat flux in nucleate boiling. 8

(b) A heat exchanger is to be designed to condense 8 kg/s of an organic liquid ($t_{sat} = 80$ °C; $h_{fg} = 600$ kJ/kg) with cooling water available at 15 °C and at a flow rate of 60 kg/s. The overall heat transfer coefficient is 480 w/m^2-deg. Calculate :

(i) The number of tubes required. the tubes are to be of 25 mm outer diameter, 2 mm thickness and 4.85 m length.

(ii) The number of tube passes. The velocity of the cooling water is not to exceed 2 m/s. 8

6. (a) Set up expression for logarithmic mean temperature difference in case of a counter flow heat exchanger. 8

(b) A chemical having specific heat of 3.3 kJ/kgK

$\varepsilon = 0.77$

$T_{ho} = 43$ °C

$T_{co} = 43.85$

NTU = 0.573

C = 0.875

flowing at the rate of 20,000 kg/h enters a parallel flow heat exchanger at 120 °C. The flow rate of cooling water is 50,000 kg/h with an inlet temperature of 20 °C. The heat transfer area is 10 m^2 and the overall heat transfer coefficient is 1050 w/m^2K. Find (i) the effectiveness of the heat exchanger (ii) the outlet temperature of water and chemical. 8

7. (a) Differentiate between molecular diffusion and eddy diffusion with some examples. 4

(b) Explain Fick's law of diffusion. What is mass diffusivity? What is its dimension? L^2T^{-1} 4

(c) For boundary layer flow over a flat plate, how is the mass transfer coefficient related to heat transfer coefficient and skin friction coefficient? 8

8. Write short notes on the following:- 4x4=16

(a) Forced convection boiling inside a tube

(b) Fouling Factor

(c) Molecular diffusion from an evaporating fluid surface.

(d) Lewis & Schmidt number.

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BE IV (ME)

3 GDGT

FINAL B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(MAIN/ATKT/EX)

(VIII SEMESTER)

ME 452 A : GAS DYNAMICS & GAS TURBINES

(M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Derive the following relations stating the assumptions used 4

A

$$\frac{T_0}{T} = 1 + \left(\frac{r-1}{2}\right) M^2$$

(Contd.)

8.2.8

(b) Air ($C_p = 1.05$ kJ/kg K, $r = 1.38$) at 3 bar, 500 k flows with a velocity of 200 m/s in a 30 cm diameter duct.

$A_1 = 0.0700 m^2$

Calculate : -

$\rho_1 = 2.070 kg/m^3$

(a) Mass flow rate, $m = 29 kg/sec$

(b) Stagnation temperature, $M = 0.445$

(c) Mach number, and $T_0 = 519 k$

(d) Stagnation pressure $P_0 = 3.417$ bar 8

8.4.7 (e) A reservoir whose temperature can be varied at a constant pressure of 1.5 bar. The air is expanded

$\frac{T_0}{T} = 1.119$

isentropically to an exit pressure 1.015 bar. Determine the temperature and mach number to produce 100 m/s velocity at exit $M = 0.77$, $T = 41.97 k$ 4

2. (a) Draw a Rayleigh line an T-S plane. Show the following on this line.

(a) Stagnation temperature lines for both subsonic and supersonic flows.

(b) Maximum stagnation temperature point

(c) Constant static and stagnation pressure lines. 8

(b) The mach number at the exit of a combustion chamber is 0.9. The ratio of stagnation temperatures at exit and entry is 3.74. If the pressure and temperature of the gas at exit are 2.5 bar and 1000°C. Determine

(a) Mach number, pressure and temperature of the gas at entry,

(b) The heat supplied per kg of the gas, and

(c) The maximum heat that can be supplied Use $r=1.3$, $C_p=1.218$ kJ/kg k 8

3. (a) Draw the schematic diagram of a simple cycle with inter cooled and heat exchanger in gas turbine and explain the working principle. Draw also the P-V and T-S diagrams of the cycle. 10

(b) A gas turbine operates at 30 kg/s mass flow. Air enters the compressor at 1 bar, 15°C and is discharged from the compressor at 10.5 bar. Combustion occurs at constant pressure with temperature rise 420 k. If the flow leaves the turbine at 1.2 bar. Determine the net power output and also the thermal efficiency use ($C_p=1.005$ kJ/kg K, $r=1.4$) 6

$\frac{T_2}{T_1}=1.95$
$T_2=562k$
$W_c=276$
$Q_{in}=422$
$T_4=507k$
$W_T=482$
$P=6.18$ MW
$P=6183.7$ kW
$W_{net}=206$
$\eta=48$

4. (a) Define polytropic efficiency. Derive suitable expressions for polytropic efficiency and bring out the relation between the polytropic efficiency and isentropic efficiency. 8

Ex 6.2

(b) In a gas turbine plant air enters the compressor at 1 bar, 7°C. It is compressed to 4 bar with isentropic efficiency 82%. The maximum temperature at the inlet to the

$T_2=415.5k$, $T_2=445k$

turbine is 800°C. The isentropic efficiency of the turbine is 85%. The calorific value of fuel used is 43.1 MJ/kg. The heat losses are 15% to the calorific value. Calculate the following

(i) Compressor work (kJ/kg) 173.915
(ii) Heat Supplied (kJ/kg) 658.7 , T_4 = 775k
(iii) Turbine work (kJ/kg) 312.5
(iv) Net work (kJ/kg) 138.64
(v) Thermal efficiency 21.04
(vi) Air fuel ratio 43
(vii) Specific fuel consumption (kg/kw.h)
(viii) Ratio of compressor to turbine work. 0.55 8

5. (a) With a neat sketch and T-S diagram, Explain the working of turbojet engine and Also, derive the expression for the thrust developed. 10

(b) Draw neat sketch and explain the general working of solid propellent engine used in rocket engines 6

6. (a) With a suitable sketch, explain the working principle of an axial flow compressor. 8

(b) Explain briefly the following performance parameter related to centrifugal compressor

(i) Power input factor.
(ii) Pressure coefficients, and
(iii) Compressor efficiency 8

7. Write short note on the following (Any four)

(a) Comparison between centrifugal and Axial flow compressor.

(b) Fanno line and Rayleigh line

(c) Surging and choking in centrifugal compressor

(d) Lift and Drag coefficient

(e) Combustion chamber requirement in Gas Turbine

(f) Thrust and specific thrust of Turbojet Engine. 16

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B.E. IV (ME)

3 **PG**

FINAL B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(VIII SEMESTER)

(MAIN/ATKT/EX)

ME - 453 A: POWER GENERATION (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Why are most of the energies converted into electricity? What is doubling time if annual increase @8% in power production? Compare power plants on the basis of capital investment, fuel cost, operational and maintenance costs for per unit power generation. 8

(b) The incremental fuel costs for two power generating units 'A' and 'B' in power plant are given by relations:

$dF_a/dP_a = 0.065\,P_a + 25$ and

$dF_b/dP_b = 0.08\,P_a + 20$,

Where, F is fuel cost in Rs/h and P is power output in MW, estimate;

(i) the economical loading of two units when total power supply is 500 MW,

(ii) the loss in fuel cost per hour if load is equally shared by both units. 8

2. (a) How is the total installed capacity of a power plant decided? State the selection criterions and working principle of a supercritical boilers used in modern power plants and enlist the functions of a condenser in thermal power plants. 8

(b) The output of a power generating unit is 500×10^6 kWh per year and average load factor is 0.70. If annual fixed charges are Rs. 50 per kW of installed capacity and annual running charges are 5% per kWh, what is the cost per kWh of energy at the bus bar? 8

3. (a) Name different types of coal pulverizing mills. State merits of pulverised coal and basic two conditions to be satisfied to burn pulverized coal. Explain with neat sketch the working of a fluidised bed combustion process. 8

(b) The maximum load on a thermal power plant of 60 MW capacity is 50 Mw and an annual load factor of 60%. The coal consumption is 1.0 kg per unit of energy generated and cost of coal is Rs. 800 per tonne. Find (i) the annual revenue earned if energy is sold at Rs. 5 per kWh and (ii) the capacity of power plant. 8

4. (a) What do you mean by mechanical super-charging and turbo-charging? State the effect of inter-cooling in turbo-charging, how turbochargers are superior to superchargers?

Draw the fuel, air, cooling water and lubricant systems for a diesel power plant. 6

(b) A 4-cylinder, 4-stroke diesel engine develops 83.5 kW at 1800 rpm with a bsfc of 0.231 kg/kWh and air fuel ratio of 23/1. The fuel has 87% carbon and 13% hydrogen with calorific value of 43.5 MJ/kg. The coolant circulates at 0.246 kg/s and rise in temperature of 50°C & exhaust temperature of 316°C, draw the energy balance sheet for the engine. Take, R = 0.302 kJ/kgK and C_p = 1.09 kJ/kgK for exhaust gases and C_p = 1.86 kJ/kgK for steam; room temperature = 17.8°C and exhaust gas pressure = 1.013 bar. 10

5. (a) What do you mean by critical size, critical mass and logarithmic decrement of a nuclear fuel? Explain with sketch the liquid metal-cooled FBR for power generation and fuel production. State the nuclear fuel handling and waste management. 10

(b) Each fission of U^{235} yields 190 MeV of useful energy. Assuming that 85% of neutrons absorbed by U^{235} cause fission, the rest being absorbed by non-fission capture to produce and isotope U^{236}; estimate the consumption of U^{235} per day to produce 3000 MW of thermal power. 6

6. (a) What are the elements contribute to the cost of electricity? Explain with sketch the daily load duration curve and how it helps in designing a power plant? State different tariff systems to charge the consumer. 8

(b) A 2000 MW thermal power station supply power to a system having maximum and minimum demand of 1900 MW and 1200 MW respectively in a year. By assuming the load duration curve to be a straight line over the year, calculate the (i) load factor 81 (ii) capacity factor 0.77 and (iii) the reserve factor. 1.05 8

7. (a) What is the working principle and conversion efficiency of a solar photovoltaic cell? Name different types of solar cells and explain with sketch the current-voltage characteristics of a silicon solar cell. 8

(b) How much energy is available in wind stream? State different types of windmills. Explain with labelled sketch the working of a three blade horizontal wind turbine for electric power generation. 8

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BE IV (ME)

3 POM

FINAL B.E. EXAMINATION - 2025

MECHANICAL ENGINEERING

(VIII SEMESTER)

(MAIN/ATKT/EX)

ME 454 A: PRODUCTION & OPERATIONS MANAGEMENT - (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Describe a production system in terms of inputs, outputs, value addition and control feedback using a suitable example. 8

(b) What is the difference between value-added and non-value-added activities in production? How is value added at different stages of the production process? 8

2. (a) Write in details about the characteristics, design and operational control of various types of production systems. 8

(b) What are common feedback mechanisms used to monitor production performance? What role do KPIs (Key Performance Indicators) play in production control? 8

3. (a) Explain the major factors influencing the location of an industry and choice of site. Differentiate between Qualitative and Quantitative methods for selecting site. 8

(b) Describe the types of plant layouts and associated materials handling techniques. 8

4. (a) What are the key components of a transfer line system? How do they function together? Describe the different types of transfer lines and their applications. 8

(b) What are end effectors in robotic material handling? How do they influence flexibility in automation? Explain the role of machine vision and sensors in robotic material handling systems. 8

5. (a) What is assembly line balancing? Explain its significance in production systems. Give details of different methods for Line Balancing. 8

(b) What is Aggregate Production Planning (APP)? Define its purpose and significance in production and operations management. 8

6. (a) Define scheduling and sequencing in the context of production and operation management. How do they differ? Describe the role of Gantt charts in scheduling. 8

(b) Schedule 5 jobs (J1, J2, J3, J4, J5) on 3 machines (M1, M2, M3) to minimize makespan. Processing times (in hours) are given below: 8

Job	J1	J2	J3	J4	J5
M1	4	9	8	6	5
M2	5	3	2	4	6
M3	8	6	7	5	10

7. (a) Explain Critical Path Method (CPM). Given the following project activities and durations (in days), draw the network diagram, find the critical path, and calculate total project duration. 8

DACE 14

Activity	A	B	C	D	E
Predecessor	--	--	A	B	C,D
Duration	5	3	7	4	2

(b) Explain Program Evaluation and Review Technique (PERT). For an activity in PERT: Optimistic time (a) = 6 days; Most likely time (m) = 9 days and Pessimistic time (b) = 18 days. Calculate: (i) Expected time (te), (ii) Standard deviation (σ) and (iii) Probability of completing within 12 days (at Z = 0.67, P=75%). 8

10

8. Write short notes. **(Attempt any four)** 4x4=16

1. Automated Guided Vehicles (AGVs)
2. Gantt Chart
3. Group technology
4. Computer aided process planning
5. Just-in-time manufacturing
6. Agile and Lean manufacturing

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B.E. IV (ME)

3

Elective

FINAL B.E. EXAMINATION - 2025
MECHANICAL ENGINEERING
(VIII SEMESTER)
(MAIN/ATKT/EX)
ELECTIVE:
ME - 455 A: FACILITIES LOCATION AND LAYOUT PLANNING (M)

Time - Three Hours
Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Explain the key factors affecting plant location and site selection? 8

(b) A multinational company plans to setup a new production unit in India. How do infrastructure and government policies influence their decision in choosing an ideal location? Provide real life examples? 8

2. (a) You are designing a production facility for a consumer electronics company would you choose process layout or product layout? Justify your choice with real life examples and key considerations. 8

May 23, 2025, 16:01

(b) A company manufacturing customized luxury cars follow a different production approach compared to a smart phone manufacturer. Explain how job shop, mass production, and special product manufacturing influence plant layout decisions. 8

3. (a) A new e-commerce fulfillment centre is being designed to handle thousands of orders daily. Discuss the key factors affecting its plant layout, including materials, machinery, movement and safety to ensure efficiency? 8

(b) A pharmacentical company is expanding its operations and needs an efficient warehouse system for storing raw materials and finished products. Explain the importance of storage and warehouse planning in plant layout for such an industry? 8

4. (a) Describe different tools used for layout development such as process charts, flow diagrams, and assembly charts? 8

(b) Exlain modern computer assisted layout development techniques such as CRAFT, CORELAP and ALDEP. 8

5. (a) A new food processing plant is being setup to produce package snacks, outline the step-by-step procedure for installing and evaluating its plant layout to ensure efficiency and hygiene. Compliance? 8

(b) An Airport needs to improve the movement of heavy loads. Describe cargo terminal different types of material handling equipment suitable for this scenario, such as mobile equipment conveyors, hoists etc. 8

6. (a) A global car manufacturing is setting up a new product facility. Discuss the role of plant engineering in ensuring an efficient installations of the plant layout and minimizing downtime. Provide real life examples? 8

(b) Explain the principles of material handling and how they can be applied to optimize the facility's plant layout? 8

7. Write short notes (any four) 4X4=16

(a) Role of automation in modern plants

(b) Principles of plant layout

(c) Group layout

(d) Storage space requirements

(e) Difference between process and product layouts

(f) Safety measures in material handling operations.